PHYS 2022 midterm examination cheatsheet

identities · equation results by lecture · methods · derivation steps

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Part 1: Trigonometric and hyperbolic identities

Trigonometric identities

I1 $\sin^2\alpha + \cos^2\alpha = 1$ Pythagorean identity; recovers either function from the other

I2 $\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$ double angle for cosine; rewrites cos^2 in integrals, the ground-state ⟨x²⟩ split uses it

I3 $\sin 2\alpha = 2\sin\alpha\cos\alpha$ double angle for sine

I4 $\sin^2\alpha = \frac{1-\cos 2\alpha}{2}, \cos^2\alpha = \frac{1+\cos 2\alpha}{2}$ half-angle powers; the ground-state ⟨x²⟩ split uses these

I5 $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$ sine in half angles; the Rutherford dσ/dΩ chain cancels sin theta against two half-angle factors

I6 $\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\cos\theta}{2}}$ half-angle radical; inverts cos theta in the Rutherford b(theta) chain

I7 $\cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$ product as a single sine; evaluates the cos theta sin theta hemisphere integral

Averages and sums

I8 $\langle\cos^2\theta\rangle_{\text{hemisphere}} = \frac{1}{3}$ isotropy average; photon-gas radiation pressure p = u/3 and the kinetic-theory pressure

I9 $\int_0^L \sin^2\left(n\pi\frac{x}{L}\right)\mathrm{d}x = \frac{L}{2}$ box normalisation and every ⟨sin^2⟩ average

I10 $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}, \sum_{n=0}^{\infty} nx^n = \frac{x}{(1-x)^2}$ for $|x| < 1$ Planck's mean-energy sum collapses through the geometric series and its derivative

Euler's formula

I11 $e^{i\alpha} = \cos\alpha + i\sin\alpha, \cos\alpha = \frac{e^{i\alpha}+e^{-i\alpha}}{2}, \sin\alpha = \frac{e^{i\alpha}-e^{-i\alpha}}{2i}$ complex exponentials and trig functions carry the same information; box and finite-well solution spaces swap between them

I12 $e^{\pm ikx} = \cos kx \pm i\sin kx$ the wave-function form used in the box, the finite well, and the parity rewrites

Hyperbolic identities

I13 $\sinh x = \frac{e^x-e^{-x}}{2}, \cosh x = \frac{e^x+e^{-x}}{2}$ hyperbolic form of the barrier interior and free-region solutions

I14 $\cosh^2 x - \sinh^2 x = 1$ Pythagorean hyperbolic identity

I15 $e^{\kappa L} - e^{-\kappa L} = 2\sinh(\kappa L),\ e^{\kappa L} + e^{-\kappa L} = 2\cosh(\kappa L)$ the tunnelling T denominator reduces through these

Applications of the identities

T1 $\langle g\rangle = \int g(v_x)f(v_x)\,\mathrm{d}v_x$ with $\int f(v_x)\,\mathrm{d}v_x = 1$ momentum integral definition

T2 $\langle v^2\rangle = 3\langle v_x^2\rangle$ isotropy step of pressure chain

T3 $\frac{1}{E} = \frac{\lambda}{hc}$ energy-to-wavelength conversion step

T5 $k = \frac{2\pi}{\lambda}$ mode-count change of variable

T6 $\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\frac{\pi}{2}}\cos\theta\sin\theta\,\mathrm{d}\theta = 2\pi\times\left(\frac{1}{2}\right) = \pi$ factor π in $I_\lambda = \pi B_\lambda$

T8 $f = \frac{c}{\lambda}$ turns hf into $\frac{hc}{\lambda}$ in every spectrum formula

T9 $E = \frac{hc}{\lambda}$ wavelength chain

T11 $E = K + V$ with $V = -\frac{e^2}{4\pi\epsilon_0 r}$ planetary-energy and ladder chains

T12 de Broglie $\lambda = \frac{h}{p}$ all three diffraction experiments and the orbit chain

T13 $\Delta k \cdot \Delta x \geq \frac{1}{2}$ with $p = \hbar k$ wave-packet bound chain

T14 $p = \hbar k$ turns $\Delta k \cdot \Delta x \geq \frac{1}{2}$ into $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$ localisation bound chain

T16 de Broglie $\lambda = \frac{h}{p}$ closes the standing-wave quantisation loop standing-wave route

T17 $\hat{p}e^{ikx} = -i\hbar(ik)e^{ikx} = \hbar k e^{ikx}$ momentum-representation derivation

T18 $E = \frac{p^2}{2}m + V$ survives as operator identity $\hat{H} = \hat{K} + \hat{V}$ operator construction

T21 Taylor expansion with $V_1 = 0$ at minimum harmonic approximation

Part 2: Constants and notation

Constants and notation

C1 $\hbar = 1.0546 \times 10^{-34}\ \mathrm{J \cdot s}$ reduced Planck constant

C2 $\mathrm{e} = 1.602 \times 10^{-19}\ \mathrm{C}$ elementary charge

C3 $\epsilon_0 = 8.854 \times 10^{-12}\,\frac{\mathrm{F}}{\mathrm{m}}$ vacuum permittivity

C4 $k_B = 1.381 \times 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$ Boltzmann constant

C5 $c = 2.998 \times 10^{8}\,\frac{\mathrm{m}}{\mathrm{s}}$ speed of light in vacuum

C6 $a_0 = 5.29 \times 10^{-11}\ \mathrm{m}$ Bohr radius

C7 $m_e = 9.109 \times 10^{-31}\ \mathrm{kg}$ electron rest mass

C8 $\sigma = 5.67 \times 10^{-8}\,\frac{\mathrm{W}}{\mathrm{m^2 \cdot K^4}}$ Stefan–Boltzmann constant

C9 $R_H = 1.097 \times 10^{7}\mathrm{m}^{-1}$ Rydberg constant for hydrogen

C10 $N_A = 6.022 \times 10^{23}\mathrm{mol}^{-1}$ Avogadro constant

C11 $h = 6.626 \times 10^{-34}\ \mathrm{J \cdot s}$ Planck constant

C12 $m_p = 1.673 \times 10^{-27}\ \mathrm{kg}$ proton rest mass

Notation: θ scattering angle, b impact parameter, K kinematic factor or kinetic energy as labelled, λ wavelength, ν frequency, u energy density, T temperature or transmission as labelled, Z_1, Z_2 atomic numbers of projectile and target.

Week 1 lecture 1

what physics is; conservation laws; fundamental forces

physics

A1 $PV = Nk_BT$ ideal-gas equation from balloon measurements *(physics)*

balloon gas measurements (physics): vary T at fixed P, repeat at other P, N
Measure: volume V. Determines: ideal-gas relation $PV = Nk_BT$.

classical physics

A2 $F = ma$ Newton's second law *(classical physics)*

A3 $F_1 = F_2 = G\left(\frac{m_1 m_2}{r^2}\right)$ Newton's law of gravitation *(classical physics)*

A4 $\nabla \cdot D = \rho_v$ Maxwell Gauss law *(classical physics)*

A5 $\nabla \cdot B = 0$ Maxwell no-monopole law *(classical physics)*

A6 $\nabla \times E = -\left(\frac{\partial B}{\partial t}\right)$ Maxwell Faraday law *(classical physics)*

A7 $\nabla \times H = \left(\frac{\partial D}{\partial t}\right) + J$ Maxwell Ampère-Maxwell law *(classical physics)*

A8 first law $\left(\frac{\mathrm{d}E}{\mathrm{d}t}\right) = 0$ energy of isolated system constant *(classical physics)*

A9 second law $\left(\frac{\mathrm{d}S}{\mathrm{d}t}\right) \geq 0$ entropy never decreases *(classical physics)*

conservation law

A10 four conserved quantities: energy, charge, linear momentum, angular momentum classical conservation laws *(conservation law)*

A11 energy ↔ time invariance, momentum ↔ space invariance, angular momentum ↔ rotation invariance, charge ↔ gauge invariance Noether-style pairings *(conservation law)*

fundamental interaction

A12 $(F)_g = -G\left(\frac{m_1 m_2}{r^2}\right)\hat{r}$ gravitational force between masses *(fundamental interaction)*

A13 $(F)_C = \left(\frac{1}{4\pi\epsilon_0}\right)\left(\frac{q_1 q_2}{r^2}\right)\hat{r}$ Coulomb force between charges *(fundamental interaction)*

A14 $F_e \propto \left(\frac{q_1 q_2}{d^2}\right)$ Coulomb's 1785 conclusion *(fundamental interaction)*

A15 fit $\frac{1}{r^{1.957}}$ beats $\frac{1}{r^2}$ on three points data do not select inverse square *(fundamental interaction)*

Coulomb torsion balance (1785) (fundamental interaction): deflect pith balls and read micrometer angles
Measure: angles of deflection. Determines: force law $\frac{F_e \propto q_1 q_2}{d^2}$.

Week 1 lecture 2

waves and particles of light; the electromagnetic spectrum; atomic theory; kinetic theory of gases

wave

A16 $\lambda = \frac{h}{p} = \frac{h}{mv}$ de Broglie wavelength of a particle *(wave)*

A17 $\frac{\Delta x \approx \lambda L}{d}$ double-slit fringe spacing, L screen distance *(wave)*

A18 $u = \frac{\Delta\omega}{\Delta k}$; $u = \frac{\mathrm{d}\omega}{\mathrm{d}k}$ group velocity of envelope *(wave)*

A19 $v_{\mathrm{ph}} = \frac{\omega}{k}$ phase velocity of crests *(wave)*

A20 $\omega = \alpha k$ non-dispersive condition, $u = v_{\mathrm{ph}} = \alpha$ *(wave)*

single-particle double slit (wave): fire particles one at a time through two slits
Measure: accumulated fringe pattern. Determines: wave behaviour of matter.

light

A21 $d \sin\theta = m\lambda$ for integer m bright-band condition of double-slit pattern *(light)*

A22 $\sin\theta_1 = \frac{\lambda}{d}$ first side band angle *(light)*

A23 $a \sin\theta = m\lambda$ for nonzero integer m single-slit envelope minima *(light)*

double-slit interference (light): pass light through two narrow slits in 1801
Measure: band positions. Determines: wave nature of light, wavelength.

electromagnetic radiation

A24 $c = \left(\frac{1}{\sqrt{\mu_0\epsilon_0}}\right)$ speed of light from permeability and permittivity *(electromagnetic radiation)*

A25 $c = \lambda f$ speed, wavelength, frequency relation *(electromagnetic radiation)*

A26 $c = 3 \times 10^8\ \mathrm{m/s}$ speed of light in vacuum *(electromagnetic radiation)*

A27 λ from 10 m to 10^{-13} m, $f = \frac{c}{\lambda}$ from 3×10^7 Hz to 3×10^{21} Hz span of the electromagnetic spectrum *(electromagnetic radiation)*

A28 radio and TV, microwave, infrared, visible light, ultraviolet, X-rays, gamma rays bands in order of decreasing wavelength *(electromagnetic radiation)*

A29 visible light 700 nm to 400 nm, red to violet visible band limits *(electromagnetic radiation)*

A30 $(F) = -e(E)$ force driving antenna current *(electromagnetic radiation)*

A31 $\lambda = \frac{c}{f}$ wavelength of wave from oscillating charge *(electromagnetic radiation)*

antenna reception (electromagnetic radiation): wave's electric field drives free electrons in receiving antenna
Measure: current at wave frequency. Determines: detection of electromagnetic waves.

history of atomic theory

A32 $N_A = 6.022 \times 10^{23}$ per mole Avogadro's constant *(history of atomic theory)*

A33 $+e$ discrete positive-charge units Thomson 1910 refinement *(history of atomic theory)*

A34 classical collapse estimate 10^{-11} s planetary-atom hidden failure *(history of atomic theory)*

A35 about 1 in 8,000 alpha particles came straight back gold-foil backscattering rate *(history of atomic theory)*

kinetic theory of gases

A36 $PV = Nk_BT = nN_Ak_BT = nRT$ ideal gas equation *(kinetic theory of gases)*

A37 $k_B = 1.38 \times 10^{-23}\ \mathrm{J/K}$ Boltzmann constant *(kinetic theory of gases)*

A38 $N_A = 6.02 \times 10^{23}\mathrm{mol}^{-1}$ Avogadro's number *(kinetic theory of gases)*

A39 $R = 8.31\mathrm{J\ mol^{-1}K^{-1}}$ ideal gas constant *(kinetic theory of gases)*

A40 $P = \left(\frac{1}{A}\right)\left(\frac{\mathrm{d}p}{\mathrm{d}t}\right)$ pressure as momentum delivered per area per time *(kinetic theory of gases)*

A41 $\mathrm{d}p = \int 2mv_x \cdot \left(\frac{1}{2}\right)nf(v_x)v_x A\,\mathrm{d}t\,\mathrm{d}v_x = nmA\langle v_x^2\rangle\,\mathrm{d}t$ momentum delivered in time dt *(kinetic theory of gases)*

A42 $P = nm\langle v_x^2\rangle$ pressure from mean square velocity *(kinetic theory of gases)*

A43 $\left(\frac{1}{2}\right)m\langle v_x^2\rangle = \left(\frac{1}{2}\right)k_BT$, so $\langle v_x^2\rangle = \frac{k_BT}{m}$ equipartition for one component *(kinetic theory of gases)*

A44 $\langle v^2\rangle = \left(\frac{3k_BT}{m}\right)$, $P = \left(\frac{N}{V}\right)k_BT$ isotropy result and ideal gas law *(kinetic theory of gases)*

A45 $P_1V_1 = P_2V_2$ Boyle's law at constant T, n *(kinetic theory of gases)*

A46 $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ Charles's law at constant P, n *(kinetic theory of gases)*

A47 $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ Gay-Lussac's law at constant V, n *(kinetic theory of gases)*

A48 $\frac{V_1}{n_1} = \frac{V_2}{n_2}$ Avogadro's law at constant P, T *(kinetic theory of gases)*

A49 $U = N|K| = nN_A|K|$ internal energy from mean kinetic energy *(kinetic theory of gases)*

A50 $|K| = \left(\frac{f}{2}\right)k_BT$ equipartition average kinetic energy *(kinetic theory of gases)*

A51 $U = \left(\frac{f}{2}\right)nRT$ internal energy of gas *(kinetic theory of gases)*

A52 $C = \frac{Q}{\Delta T} = \frac{\Delta U}{\Delta T}$, $C_V = \frac{C}{n}$ heat capacity definitions *(kinetic theory of gases)*

A53 $C_V = \left(\frac{\mathrm{d}U}{n\,\mathrm{d}T}\right) = \left(\frac{fR}{2}\right)$ molar heat capacity at constant volume *(kinetic theory of gases)*

A54 $C_p = C_V + R$ heat capacity at constant pressure *(kinetic theory of gases)*

A55 $\frac{c_V}{R}$ plateaus at $\frac{3}{2}$, $\frac{5}{2}$, $\frac{7}{2}$ translation, rotation, vibration activation *(kinetic theory of gases)*

A56 $C_p = C_V + R$ levels: $f = 3$ noble gases, $f = 5$ air, $f = 6$ hydrogen, $f = 13$ carbon dioxide over 300 to 3500 K constant-pressure degrees of freedom *(kinetic theory of gases)*

A57 $\frac{\mathrm{Rate}_1}{\mathrm{Rate}_2} = \sqrt{\frac{m_2}{m_1}}$ Graham's law of effusion *(kinetic theory of gases)*

A58 $v_{\mathrm{rms}} = \sqrt{\frac{3k_BT}{m}}$ root-mean-square molecular speed *(kinetic theory of gases)*

A59 helium $C_V = 12.5$, argon 12.6 with $f = 3$; nitrogen 20.7, oxygen 20.8 with $f = 5$; ammonia 29.1, carbon dioxide 29.7 with $f = 6$ measured molar heat capacities *(kinetic theory of gases)*

A60 alpha helix about $800\mathrm{cal\ mol^{-1}K^{-1}} = 400R$, $f = 800$; chymotrypsinogen about $9000\mathrm{cal\ mol^{-1}K^{-1}} = 4500R$, $f = 9000$ large-molecule heat capacities *(kinetic theory of gases)*

wall-force measurement (kinetic theory of gases): measure momentum delivered to wall of area A in time dt
Measure: $\frac{\mathrm{d}p}{\mathrm{d}t}$. Determines: gas pressure, test of $PV = Nk_BT$.

heat-capacity measurement (kinetic theory of gases): measure molar heat capacity C_V of the gas
Measure: heat added per temperature change. Determines: number of degrees of freedom f.

effusion timing (kinetic theory of gases): let gas effuse through tiny hole into vacuum
Measure: effusion rate. Determines: molar mass through Graham's law.

Derivation chains

apply equipartition

{chain} 1 $\left(\frac{1}{2}\right)m\langle v_x^2\rangle = \left(\frac{1}{2}\right)k_BT$ gives $\langle v_x^2\rangle = \frac{k_BT}{m}$ | component mean square

assemble the gas law

{chain} 1 $P = \left(\frac{N}{V}\right)k_BT$ rearranges to $PV = Nk_BT$ | ideal gas equation

count molecules hitting

{chain} 1 those with v_x in $[v_x, v_x + \mathrm{d}v_x]$ hit within $\mathrm{d}t$ only from slab of thickness $v_x\,\mathrm{d}t$; count is $\left(\frac{1}{2}\right) n f(v_x)\,\mathrm{d}v_x \cdot A \cdot v_x\,\mathrm{d}t$ | molecules in slab

define pressure

{chain} 1 $P = \left(\frac{1}{A}\right)\left(\frac{\mathrm{d}p}{\mathrm{d}t}\right)$, momentum delivered per unit area per unit time | macroscopic pressure

divide by area and time

{chain} 1 $P = nm\langle v_x^2 \rangle$ | pressure formula

integrate over speeds

{chain} 1 $\mathrm{d}p = \int 2mv_x \cdot \left(\frac{1}{2}\right) n f(v_x) v_x A\,\mathrm{d}t\,\mathrm{d}v_x = nmA\langle v_x^2 \rangle\,\mathrm{d}t$ | momentum in time $\mathrm{d}t$

momentum per collision

{chain} 1 each collision reverses v_x and delivers $\Delta p = 2mv_x$ | momentum transfer

use isotropy

{chain} 1 $\langle v^2 \rangle = 3\langle v_x^2 \rangle$, so $\langle v^2 \rangle = \left(\frac{3k_B T}{m}\right)$ | mean square speed

Week 2 lecture 1

the electron; the photoelectric effect; X-rays; the Compton effect

photoelectric effect

A61 $KE_{\max} = eV_s$ maximum kinetic energy from stopping potential *(photoelectric effect)*

A62 work functions: Na ≈ 2.3 eV, Zn 4.3 eV, Cu 4.7 eV, Pt 6.35 eV typical metal work functions *(photoelectric effect)*

A63 $E = hf$ photon energy *(photoelectric effect)*

A64 $hf = \Phi + K$ photoelectric energy balance *(photoelectric effect)*

A65 $K = hf - \Phi$, $KE_{\max} = hf - \Phi$ photoelectric equation *(photoelectric effect)*

A66 $f_0 = \frac{\Phi}{h}$ threshold frequency *(photoelectric effect)*

A67 $V_s = \left(\frac{h}{e}\right) f - \frac{\Phi}{e}$ stopping voltage straight line in frequency *(photoelectric effect)*

A68 slope $\frac{h}{e}$ universal, intercept gives Φ line-shape facts *(photoelectric effect)*

A69 CCD quantum efficiency ≈ 70% visible typical CCD photon-to-electron yield *(photoelectric effect)*

photoelectric measurement (photoelectric effect): vacuum tube with emitter and collector; retarding voltage, ammeter
Measure: electron current vs retarding voltage. Determines: stopping potential V_s, then KE_max = eV_s and h, Φ from V_s versus f.

X-ray

A70 $hf = E_i - E_f$ bremsstrahlung photon energy *(X-ray)*

A71 $hf_{\max} = eV$ maximum photon energy from accelerating voltage *(X-ray)*

A72 30 to 150 kV; 99% heat, 1% X-rays X-ray tube operating numbers *(X-ray)*

A73 K: $n = 1$; L: $n = 2$; $K\alpha$: L→K; $K\beta$: M→K; $L\alpha$: M→L; $L\beta$: N→L characteristic line naming *(X-ray)*

A74 $\left(\frac{1}{\lambda_K}\right) = R(Z-1)^2\left(\left(\frac{1}{1^2}\right) - \left(\frac{1}{n^2}\right)\right)$, $n = 2, 3, \ldots$ Moseley K-series law *(X-ray)*

A75 $f_{K\alpha} = \left(\frac{3}{4}\right) cR(Z-1)^2$ K-alpha frequency *(X-ray)*

A76 $\sqrt{f} = A(Z-b)$, $b \approx 1$ Moseley general form *(X-ray)*

A77 effective charge $(Z-1)e$ screened nuclear charge felt by K electron *(X-ray)*

A78 Wien/Sommerfeld 10^{-10} to 10^{-11} m; lattice 10^{-10} m X-ray and crystal scales *(X-ray)*

Moseley X-ray method (X-ray): measure characteristic X-ray frequencies across the elements
Measure: frequency f of K-series lines vs Z. Determines: atomic number Z, shell structure with screening constant b.

Compton effect

A79 $p = \frac{E}{c} = \frac{h}{\lambda}$ photon momentum *(Compton effect)*

A80 $E_e^2 = \left(mc^2\right)^2 + p_e^2 c^2$ relativistic electron energy-momentum relation *(Compton effect)*

A81 $E_\gamma + mc^2 = E_\gamma{}' + E_e$, $(p)_\gamma = (p)_\gamma{}' + (p)_e$ energy and momentum conservation *(Compton effect)*

A82 $p_e^2 = p_\gamma^2 + p_\gamma{}'^2 - 2p_\gamma p_\gamma{}' \cos\theta$ squared momentum law *(Compton effect)*

A83 $E_e^2 = E_\gamma^2 + E_\gamma{}'^2 + \left(mc^2\right)^2 - 2E_\gamma E_\gamma{}' + 2mc^2\left(E_\gamma - E_\gamma{}'\right)$ squared energy law *(Compton effect)*

A84 $E_\gamma E_\gamma{}'(1 - \cos\theta) = mc^2\left(E_\gamma - E_\gamma{}'\right)$ combined conservation result *(Compton effect)*

A85 $1 - \cos\theta = mc^2\left(\frac{\lambda' - \lambda}{hc}\right)$ wavelength form of the shift *(Compton effect)*

A86 $\Delta\lambda = \lambda' - \lambda = \left(\frac{h}{mc}\right)(1 - \cos\theta)$ Compton shift formula *(Compton effect)*

A87 $\lambda_C = \frac{h}{m_e c} \approx 2.43 \times 10^{-12}$ m electron Compton wavelength *(Compton effect)*

A88 $\lambda_C = \frac{h}{m_p c} \approx 1.32 \times 10^{-15}$ m proton Compton wavelength *(Compton effect)*

A89 shift λ_C at 90°, $2\lambda_C$ at 180° angle extremes of the shift *(Compton effect)*

Compton scattering (Compton effect): crystal spectrometer and ionisation chamber measure X-rays off graphite at several angles
Measure: scattered wavelength against scattering angle. Determines: wavelength shift vs θ, evidence of photon momentum.

Derivation chains

Compton shift chain

{chain} 1 struck particle at rest: $E_\gamma + mc^2 = E_\gamma{}' + E_e$ and $(p)_\gamma = (p)_\gamma{}' + (p)_e$

{chain} 2 rearrange for $(p)_e$ and square: cross term $2(p)_\gamma \cdot (p)_\gamma{}' = 2p_\gamma p_\gamma{}' \cos\theta$ leaves $p_e^2 = p_\gamma^2 + p_\gamma{}'^2 - 2p_\gamma p_\gamma{}' \cos\theta$

{chain} 3 rearrange for E_e and square: $E_e^2 = E_\gamma^2 + E_\gamma{}'^2 + \left(mc^2\right)^2 - 2E_\gamma E_\gamma{}' + 2mc^2\left(E_\gamma - E_\gamma{}'\right)$

{chain} 4 substitute $p_e^2 c^2 = E_e^2 - \left(mc^2\right)^2$ into the squared laws; E_γ^2, $E_\gamma{}'^2$, $\left(mc^2\right)^2$ cancel, leaving $E_\gamma E_\gamma{}'(1 - \cos\theta) = mc^2\left(E_\gamma - E_\gamma{}'\right)$

{chain} 5 divide by $E_\gamma E_\gamma{}'$ and use $\frac{1}{E} = \frac{\lambda}{hc}$ on the right: $1 - \cos\theta = mc^2\left(\frac{\lambda' - \lambda}{hc}\right)$

{chain} 6 $\Delta\lambda = \lambda' - \lambda = \left(\frac{h}{mc}\right)(1 - \cos\theta)$; zero at θ = 0, doubled at backscattering

Moseley (Z–1)² screening chain

{chain} 1 one K-shell electron remains beside the ejected one and screens the nucleus, so the K electron sees $(Z-1)e$

{chain} 2 Bohr energies scale with the square of the seen charge, so a drop into the K shell radiates $f \propto (Z-1)^2$

{chain} 3 $\left(\frac{1}{\lambda_K}\right) = R(Z-1)^2\left(\left(\frac{1}{1^2}\right) - \left(\frac{1}{n^2}\right)\right)$; the $n = 2$ to $n = 1$ drop gives $f_{K\alpha} = \left(\frac{3}{4}\right) cR(Z-1)^2$

{chain} 4 general form $\sqrt{f} = A(Z-b)$ collects the level difference into A with screening $b \approx 1$, so $\sqrt{f}$ against Z is a line

photoelectric energy balance

{chain} 1 one photon transfers all its energy in one collision: $hf = \Phi + K$, work function paid, remainder kinetic

{chain} 2 most energetic electrons leave the surface with no travel cost, so $KE_{\max} = hf - \Phi$

{chain} 3 below $f_0 = \frac{\Phi}{h}$ no emission: $hf < \Phi$

{chain} 4 with $KE_{\max} = eV_s$: $V_s = \left(\frac{h}{e}\right) f - \frac{\Phi}{e}$, slope $\frac{h}{e}$ shared by all metals, intercept $-\frac{\Phi}{e}$

Week 2 lecture 2

line spectra and the Balmer formula; cavity black bodies; the Wien, Rayleigh–Jeans, and Planck laws; the positron and annihilation

emission spectrum

A90 $d \sin\theta = n\lambda$ with n an integer grating equation, spacing d between rulings *(emission spectrum)*

A91 $E_1 - E_0 = hf$ photon energy of a level drop *(emission spectrum)*

A92 lithium crimson, potassium pale lilac, iron orange, barium yellow-green, strontium crimson standard flame colours *(emission spectrum)*

A93 hydrogen four lines across visible, iron lines crowding into a dense band line counts of compared spectra *(emission spectrum)*

A94 galaxy behind SMACS 0723, light left 13.1 billion years ago, spectrum over rough astronomical spectrum example *(emission spectrum)*

A95 detector gap near 4.3μ m instrument blindness in that spectrum *(emission spectrum)*

flame test with spectrometer (emission spectrum): dip wire in salt, hold in flame, read spectrum
Measure: line wavelengths. Determines: element identity.

grating spectroscopy (emission spectrum): collimate beam, spread by grating onto detector
Measure: diffraction angle θ. Determines: wavelength of each line.

astronomical emission spectroscopy (emission spectrum): microshutter array spectrometer over 3.3 to 4.9μ m
Measure: line positions. Determines: composition of remote galaxy.

Balmer series

A96 $\lambda = 364.56\left(\frac{k^2}{k^2-4}\right)$ nm with $k = 3, 4, 5, \ldots$ and $k > 2$ Balmer formula *(Balmer series)*

A97 $k = 3$: $\lambda = 364.6 \times \left(\frac{9}{5}\right) \approx 656.3$ nm red line *(Balmer series)*

A98 $k = 4$: $\lambda = 364.56 \times \left(\frac{4}{3}\right) = 486.1$ nm blue-green line *(Balmer series)*

A99 $k = 5$ and $k = 6$: 434.0 nm and 410.2 nm violet lines *(Balmer series)*

A100 $k = 7$: 397.0 nm; 365.0 nm among lines near limit ultraviolet run of the series *(Balmer series)*

A101 series limit 364.56 nm shortest Balmer wavelength *(Balmer series)*

A102 $E_n = -13.6\frac{\text{eV}}{n^2}$ Bohr hydrogen energy levels *(Balmer series)*

A103 $\frac{hc}{\lambda} = E_0\left(\frac{1}{4} - \frac{1}{k^2}\right)$ with $E_0 = 13.6$ eV photon energy of drop to second level *(Balmer series)*

A104 $\lambda = \left(\frac{4hc}{E_0}\right)\frac{k^2}{k^2-4}$ Bohr wavelength formula, Balmer shape *(Balmer series)*

A105 $\frac{4hc}{E_0 \approx 364.6}$ nm Bohr value of Balmer constant *(Balmer series)*

plate or detector spectroscopy (Balmer series): photograph or record Balmer lines past the visible
Measure: line positions. Determines: wavelengths toward series limit.

thermal radiation

A106 $c = f\lambda$ frequency and wavelength of thermal radiation *(thermal radiation)*

A107 $\lambda_{\max} T = 2.898 \times 10^{-3}$ m · K peak position set by temperature alone *(thermal radiation)*

A108 total radiated power per unit area rises as T^4 Stefan–Boltzmann scaling *(thermal radiation)*

A109 below 800 K black, 900 K dull red, 1750 K red, 3200 K orange to yellow, 5500 K colour-temperature ladder *(thermal radiation)*

A110 eye responds to about 380 to 700 nm visible band the eye detects *(thermal radiation)*

glow-colour thermometry (thermal radiation): observe colour of glowing body
Measure: hue. Determines: coarse temperature estimate.

infrared thermometry (thermal radiation): read radiation outside the visible band
Measure: emitted intensity. Determines: body temperature without contact.

infrared

A111 wavelengths from roughly 700 nm up to about 1 mm infrared band limits *(infrared)*

A112 black-body radiator below about 800 K looks black while hot visible darkness of hot body *(infrared)*

prism-and-thermometer spectroscopy (infrared): split sunlight with prism, read thermometer in each colour
Measure: temperature reading per colour. Determines: invisible radiation past the red edge.

bolometer or semiconductor detection (infrared): respond to non-visual effect of radiation
Measure: signal in band beyond visible. Determines: radiation the eye misses.

black-body radiation

A113 $B_\lambda(T) = \left(\frac{2hc^2}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ Planck's law spectral radiance of black body *(black-body radiation)*

A114 $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda$ radiance from isotropic cavity energy density *(black-body radiation)*

A115 $I_\lambda = \pi B_\lambda$ flux from radiance, cosine-weighted hemisphere *(black-body radiation)*

A116 $p = \frac{u}{3}$ photon-gas radiation pressure on cavity walls *(black-body radiation)*

A117 $U = u(T)V$ photon-gas total energy *(black-body radiation)*

A118 $u = aT^4$ photon-gas energy density from thermodynamics *(black-body radiation)*

A119 $R = \sigma T^4$ radiated power per unit area *(black-body radiation)*

A120 $\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m \cdot K}$ Wien displacement constant, non-contact thermometer *(black-body radiation)*

A121 $\lambda_{\max} = \frac{2.898\times10^{-3}}{2.73\approx1.06\times10^{-3}}\ \mathrm{m}$ cosmic microwave background peak near 1 mm *(black-body radiation)*

cavity spectroscopy (black-body radiation): heat enclosure, read through small hole
Measure: hole brightness at one wavelength. Determines: cavity temperature.

microwave receiver fitting (black-body radiation): measure relic radiation spectrum, fit black-body curve
Measure: spectrum against wavelength. Determines: background temperature, spectrum universality.

Wien's displacement law

A122 $\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m \cdot K}$ peak wavelength in inverse proportion to temperature *(Wien's displacement law)*

A123 $5(1-e^{-x}) = x$, root $x \approx 4.9651$ peak condition on the Planck spectrum *(Wien's displacement law)*

A124 $\lambda_{\max} T = \left(\frac{hc}{4.9651 k_B}\right) = 2.898 \times 10^{-3}\ \mathrm{m \cdot K}$ displacement constant from h, c, k_B alone *(Wien's displacement law)*

A125 peak intensity rises as the fifth power of T height scaling of the black-body peak *(Wien's displacement law)*

A126 $\frac{2.898\times10^{-3}}{1800\approx1.61\times10^{-6}}\ \mathrm{m}$ 1800 K curve peaks near 1.6μ m *(Wien's displacement law)*

A127 $\frac{2.898\times10^{-3}}{310\approx9.3\mu}\ \mathrm{m}$ body at 37°C peaks in infrared *(Wien's displacement law)*

A128 $\frac{2.898\times10^{-3}}{3500\approx830}\ \mathrm{nm}$ Betelgeuse peak past red edge *(Wien's displacement law)*

A129 $\frac{2.898\times10^{-3}}{12000\approx240}\ \mathrm{nm}$ Rigel peak inside ultraviolet *(Wien's displacement law)*

non-contact thermometry (Wien's displacement law): measure peak wavelength of emitted spectrum
Measure: $\lambda_{\max}$. Determines: body temperature.

stellar peak reading (Wien's displacement law): measure peak wavelength of stellar spectrum
Measure: $\lambda_{\max}$. Determines: stellar surface temperature.

Stefan–Boltzmann law

A130 $R(T) = \int_0^\infty I_\lambda(\lambda, T)\,\mathrm{d}\lambda = \varepsilon\sigma T^4$ total radiated power per unit area *(Stefan–Boltzmann law)*

A131 $\sigma = 5.6704 \times 10^{-8}\ \frac{\mathrm{W}}{\mathrm{m^2 \cdot K^4}}$ measured Stefan–Boltzmann constant *(Stefan–Boltzmann law)*

A132 $\sigma = \left(\frac{2\pi^5 k_B^4}{15h^3c^2}\right)$ constant from h, c, k_B alone *(Stefan–Boltzmann law)*

A133 emissivity ε ratio of real surface power to black-body power, exactly 1 for ideal surface emission factor *(Stefan–Boltzmann law)*

A134 $\left(\frac{313}{310}\right)^4 - 1 \approx 0.039$ power rise of body at 40 against 37°C *(Stefan–Boltzmann law)*

A135 $(2T)^4 = 16T^4$ power multiplied by sixteen on doubling temperature *(Stefan–Boltzmann law)*

radiometry (Stefan–Boltzmann law): measure total emitted power per unit area at known temperature
Measure: R at given T. Determines: σ or emissivity ε.

Rayleigh–Jeans law

A136 $I_\lambda = \left(\frac{2\pi c k_B T}{\lambda^4}\right)$ classical flux, diverges as $\lambda \to 0$ *(Rayleigh–Jeans law)*

A137 $I_\lambda = \left(\frac{2\pi c}{\lambda^4}\right)\overline{E}$ law with mean energy written separately *(Rayleigh–Jeans law)*

A138 $\left(\frac{1}{2}\right)k_B T$ equipartition share per quadratic degree of freedom *(Rayleigh–Jeans law)*

A139 $\left(\frac{1}{2}\right)k_B T + \left(\frac{1}{2}\right)k_B T = k_B T$ mean energy of one mode *(Rayleigh–Jeans law)*

A140 $\lambda \gg h\frac{c}{k_B T} = 1.4388 \times 10^{-2}\ \mathrm{m \cdot \frac{K}{T}}$ validity condition in wavelength, 48μ m at 300 K *(Rayleigh–Jeans law)*

A141 $u_\lambda = \left(\frac{8\pi k_B T}{\lambda^4}\right)$, $8\pi k_B T \approx 1.04 \times 10^{-19}\ \mathrm{J}$ energy density prefactor at 300 K *(Rayleigh–Jeans law)*

A142 $I_\lambda \approx \left(\frac{7.80\times10^{-12}}{\lambda^4}\right)\ \mathrm{in\ W\ m^{-3}}$ numerical flux at 300 K *(Rayleigh–Jeans law)*

A143 classical exceeds full by $\left(\frac{e^x-1}{x}\right)$; full is $\left(\frac{x}{e^x-1}\right)$ of the classical value error ratios against Planck's law *(Rayleigh–Jeans law)*

A144 overestimate about 2% at 1 mm, factor of 25 at 10^{-5} m, factor of about 4.7×10^{39} at 500 nm classical errors at 300 K *(Rayleigh–Jeans law)*

A145 $E(f) = \left(\frac{hf}{e^{h\frac{f}{k_B T}}-1}\right)$ replacement mean energy from Planck's law *(Rayleigh–Jeans law)*

classical-limit check (Rayleigh–Jeans law): evaluate $x = h\frac{c}{\lambda k_B T}$ at measured wavelength
Measure: ratio x. Determines: expected size of classical error.

Wien approximation

A146 $E(f) = \varepsilon e^{-\frac{\varepsilon}{k_B T}}$ Wien mean mode energy, $\varepsilon = hf = \frac{hc}{\lambda}$ *(Wien approximation)*

A147 $E = \left(\frac{1\cdot hf e^{-h\frac{f}{k_B T}}}{1+e^{-h\frac{f}{k_B T}}}\right) = \left(\frac{hf}{e^{h\frac{f}{k_B T}}+1}\right)$ exact two-state truncated average *(Wien approximation)*

A148 $E \approx hf e^{-h\frac{f}{k_B T}}$ Wien expression for $hf \gg k_B T$ *(Wien approximation)*

A149 $u_\lambda = \left(\frac{8\pi}{\lambda^4}\right)\cdot\left(\frac{hc}{\lambda}\right)e^{-h\frac{c}{\lambda k_B T}} = \left(\frac{8\pi hc}{\lambda^5}\right)e^{-h\frac{c}{\lambda k_B T}}$ Wien energy density *(Wien approximation)*

A150 $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda = \left(\frac{2hc^2}{\lambda^5}\right)e^{-h\frac{c}{\lambda k_B T}}$ Wien spectral radiance *(Wien approximation)*

A151 $2hc^2 \approx 1.19 \times 10^{-16}\ \mathrm{J \cdot m^2 s^{-1}}$ scale constant of the law *(Wien approximation)*

A152 $\left(\frac{hc}{k_B}\right) = 1.439 \times 10^{-2}\ \mathrm{m \cdot K}$ second scale constant of the law *(Wien approximation)*

A153 $x = \left(\frac{hc}{\lambda k_B T}\right)$ quantum-to-thermal energy ratio deciding accuracy *(Wien approximation)*

A154 $I_\lambda = \left(\frac{2\pi c}{\lambda^4}\right)\overline{E}$ shared chain all three laws plug into *(Wien approximation)*

A155 $\lambda_{\max} = \left(\frac{hc}{5k_B T}\right)$ peak of the Wien law *(Wien approximation)*

A156 $2.878 \times 10^{-3}\ \mathrm{m \cdot K}$ displacement constant this law predicts, 0.7% low *(Wien approximation)*

A157 $\frac{180}{2\pi^4} = 0.924$ fraction of true total power this law gives *(Wien approximation)*

A158 $1 - e^{-x}$ fraction of the full average the truncated one reaches *(Wien approximation)*

A159 at 300 K and 1μ m: $1.7 \times 10^{-7}\ \mathrm{W\ m^{-3} sr^{-1}}$; at 1 cm: 1.19×10^{-6} against 2.48×10^{-4}, short by 209 numeric accuracy checks *(Wien approximation)*

spectrum comparison (Wien approximation): evaluate law against true spectrum at chosen wavelengths
Measure: radiance ratio. Determines: validity range of the approximation.

ultraviolet catastrophe

A160 $I_\lambda = \frac{2\pi c k_B T}{\lambda^4}$ Rayleigh–Jeans prediction, diverges as $\lambda \to 0$ *(ultraviolet catastrophe)*

A161 $B_\lambda = 2hc^2 \frac{e^{-h\frac{c}{\lambda k_B T}}}{\lambda^5}$ Wien radiance prediction, vanishes at short wavelengths *(ultraviolet catastrophe)*

A162 $n(\lambda) = \frac{8\pi}{\lambda^4}$ mode count, confirmed by direct experiment *(ultraviolet catastrophe)*

A163 $x = h\frac{c}{\lambda k_B T}$ one quantum of energy against thermal energy *(ultraviolet catastrophe)*

A164 at 500 nm and 300 K: $x \approx 96$, $e^x - 1 \approx 4.5 \times 10^{41}$, overestimate factor $\left(\frac{e^x-1}{x}\right) \approx 4.7 \times 10^{39}$ classical overestimate at short wavelength *(ultraviolet catastrophe)*

A165 at 10^{-5} m and 300 K: $x \approx 4.8$, $\frac{x}{e^x-1} \approx 0.040$ classical overestimate by factor about 25 *(ultraviolet catastrophe)*

A166 suppression about e^{-96}, a fall by a factor of 10^{41} required fall from the $k_B T$ value *(ultraviolet catastrophe)*

A167 allowed energies $0, hf, 2hf, 3hf, \ldots$ Planck's discrete oscillator ladder *(ultraviolet catastrophe)*

A168 $\left(\frac{hf}{e^{h\frac{f}{k_B T}}-1}\right)$ thermal sum over discrete energies *(ultraviolet catastrophe)*

A169 limits: $k_B T$ for $hf \ll k_B T$; $hf e^{-h\frac{f}{k_B T}}$ for $hf \gg k_B T$ mean energy meeting both constraints *(ultraviolet catastrophe)*

A170 $e^{-h\frac{c}{\lambda k_B T}}$ factor turning the $\frac{1}{\lambda^4}$ rise into a maximum *(ultraviolet catastrophe)*

mode density of a cavity

A171 $k_x = \left(\frac{\pi n_x}{L}\right)$ along the first axis, likewise k_y and k_z standing-wave condition in cubic box *(mode density of a cavity)*

A172 $\left(\frac{L}{\pi}\right)^3$ points per unit volume k-space lattice density *(mode density of a cavity)*

A173 $4\pi k^2\,\mathrm{d}k$ volume of thin spherical shell *(mode density of a cavity)*

A174 $\left(\left(\frac{L}{\pi}\right)\right)^3 4\pi k^2\,\mathrm{d}k \times \left(\frac{2}{8}\right)$ modes in the shell with octant and polarisation factors *(mode density of a cavity)*

A175 $\left(\frac{k^2\,\mathrm{d}k}{\pi^2}\right)$ mode count per unit volume *(mode density of a cavity)*

A176 $k = \frac{2\pi}{\lambda}$, $|\,\mathrm{d}k| = \left(\frac{2\pi}{\lambda^2}\right)\mathrm{d}\lambda$ change of variable to wavelength *(mode density of a cavity)*

A177 $n(\lambda) = \frac{8\pi}{\lambda^4}$ mode density per unit volume per unit wavelength *(mode density of a cavity)*

A178 $u_\lambda = n(\lambda)\overline{E}$, then $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda$, then $I_\lambda = \pi B_\lambda = \left(\frac{1}{4}\right)c u_\lambda$ chain from count to flux *(mode density of a cavity)*

A179 $\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\frac{\pi}{2}} B_\lambda \cos\theta \sin\theta\,\mathrm{d}\theta = 2\pi \times \left(\frac{1}{2}\right) = \pi$ cosine-weighted hemisphere integral *(mode density of a cavity)*

A180 energy density $8\pi\frac{\overline{E}}{\lambda^4}$, radiance $2c\frac{\overline{E}}{\lambda^4}$, flux $2\pi c\frac{\overline{E}}{\lambda^4}$ prefactors for given mean energy *(mode density of a cavity)*

A181 $I_\lambda = \left(\frac{2\pi c}{\lambda^4}\right)\overline{E}$ shared expression of all three laws *(mode density of a cavity)*

A182 $I_\lambda = \left(\frac{2\pi hc^2}{\lambda^5}\right)e^{-h\frac{c}{\lambda k_B T}}$ Wien flux from the chain *(mode density of a cavity)*

A183 $I_\lambda = \left(\frac{2\pi hc^2}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ Planck flux from the chain *(mode density of a cavity)*

A184 $\overline{E} = hf$ with $f = \frac{c}{\lambda}$ gives $\overline{E} = \frac{hc}{\lambda}$ origin of the fifth power *(mode density of a cavity)*

cavity mode counting (mode density of a cavity): count standing waves admitted by box of known size
Measure: number of modes per band. Determines: mode density, no energy assumption.

Planck's law

A185 $B_\lambda(T) = \left(\frac{2hc^2}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ standard form of Planck's law *(Planck's law)*

A186 $n(\lambda) = \frac{8\pi}{\lambda^4}$ cavity modes per unit volume per unit wavelength *(Planck's law)*

A187 $u_\lambda = \left(\frac{8\pi}{\lambda^4}\right)\cdot\left(\frac{\frac{hc}{\lambda}}{e^{h\frac{c}{\lambda k_B T}}-1}\right) = \left(\frac{8\pi hc}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ energy density per unit wavelength *(Planck's law)*

A188 $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda$ radiance from energy density *(Planck's law)*

A189 $I_\lambda = \left(\frac{1}{4}\right)cu_\lambda$ flux from energy density *(Planck's law)*

A190 $E_n = nhf$ allowed oscillator energies, n integer *(Planck's law)*

A191 $\Delta E = hf$ size of each emission or absorption step *(Planck's law)*

A192 $h = 6.6261\times10^{-34}\ \mathrm{J\cdot s}$ accepted value of Planck's constant *(Planck's law)*

A193 $E(f) = \left(\frac{hf}{e^{h\frac{f}{k_B T}}-1}\right)$ mean energy of quantised oscillator *(Planck's law)*

A194 $I_\lambda = \left(\frac{2\pi c k_B T}{\lambda^4}\right)$ Rayleigh–Jeans limit at long wavelengths *(Planck's law)*

A195 $8\pi\times hc\times\left(\frac{c}{4\pi}\right) = 2hc^2$ origin of the prefactor *(Planck's law)*

A196 $T = 2.73\ \mathrm{K}$, peak $\approx 1.06\ \mathrm{mm}$, $h\frac{c}{\lambda k_B T}\approx 4.97$, classical law high by about 29 cosmic microwave background checks *(Planck's law)*

A197 $\lambda \gg h\frac{c}{k_B T}\approx 5.3\ \mathrm{mm}$ range where Rayleigh–Jeans limit holds for background *(Planck's law)*

oscillator population weighing (Planck's law): weight level n by Boltzmann factor, sum energies over weights
Measure: Boltzmann weights $e^{-nh\frac{f}{k_B T}}$, assumed. Determines: mean mode energy.

CMB spectroscopy (Planck's law): measure background spectrum with microwave receivers, compare with curve
Measure: spectrum at each wavelength. Determines: test of black-body law.

Max Planck

A198 $h = 6.55\times10^{-27}\ \mathrm{erg\cdot s}$, that is $6.55\times10^{-34}\ \mathrm{J\cdot s}$ Planck's own first value of h *(Max Planck)*

A199 $6.62607015\times10^{-34}\ \mathrm{J\cdot s}$ modern value, exact since 2019 SI revision *(Max Planck)*

A200 about 1.2 percent Planck's value below the modern figure *(Max Planck)*

positron

A201 positron: charge $+1e$, same mass as electron, spin $\left(\frac{1}{2}\right)\hbar$ positron properties *(positron)*

A202 antiparticle pairing: same mass, opposite charge (1928, Dirac) antimatter pairing rule *(positron)*

A203 hydrogen $\leftrightarrow$ antihydrogen: proton+electron vs antiproton+positron, same mass atomic antimatter example *(positron)*

A204 $2m_e c^2 = 1.022\ \mathrm{MeV}$ pair-production threshold energy *(positron)*

A205 photon momentum $\frac{E}{c}$ recoil condition for pair production *(positron)*

A206 charge balance $+1e \to 0 + 1e + 0$ positron-emission decay balance *(positron)*

Anderson cloud chamber (1932) (positron): photograph track in magnetic field
Measure: track curvature direction and tightness. Determines: positive charge with electron mass (radius set by momentum per unit charge).

annihilation

A207 $E = \Delta mc^2$ mass-energy relation *(annihilation)*

A208 $\Delta m = 2m_e$ for pair at rest annihilation mass change *(annihilation)*

A209 $2m_e c^2 = 1.64\times10^{-13}\ \mathrm{J} = 1.022\ \mathrm{MeV}$ total annihilation energy *(annihilation)*

A210 $m_e c^2 = 8.2\times10^{-14}\ \mathrm{J} = 511\ \mathrm{keV}$ each photon's energy *(annihilation)*

A211 two photons leave 180° apart momentum-conservation geometry at rest *(annihilation)*

A212 moving pair: opening angle below 180°, still two photons tilted geometry *(annihilation)*

A213 $\lambda = \frac{hc}{E} = 2.4\times10^{-12}\ \mathrm{m}$ annihilation photon wavelength *(annihilation)*

A214 2.4 pm: about 1/40 of an atom, 2×10^5 times shorter than 500 nm wavelength comparisons *(annihilation)*

positron emission tomography

A215 two 511 keV gamma photons, exactly opposite annihilation signal PET detects *(positron emission tomography)*

A216 detector ring + intersection of lines image reconstruction principle *(positron emission tomography)*

A217 coincidence window rule both photons within short time window *(positron emission tomography)*

A218 energy window at 511 keV rejects scattered photons *(positron emission tomography)*

positron emission tomography (positron emission tomography): inject tracer with positron-emitting isotope, detect both 511 keV photons in coincidence on a ring
Measure: coincident photon pairs. Determines: location of tracer uptake via intersecting annihilation lines.

Derivation chains

equipartition share

{chain} 1 a quadratic degree of freedom carries $\left(\frac{1}{2}\right)k_B T$, independent of mass or stiffness | classical mean energy per degree of freedom

Bohr levels

{chain} 1 $E_n = -13.6\frac{\mathrm{eV}}{n^2}$ | hydrogen energy ladder

Boltzmann-weighted mean

{chain} 1 $E(f) = \left(\frac{\sum_n nhfe^{-nh\frac{f}{k_B T}}}{\sum_n e^{-nh\frac{f}{k_B T}}}\right)$ | mean oscillator energy as ratio of sums

Planck peak

{chain} 1 $B_\lambda \propto \frac{\lambda^{-5}}{e^x-1}$ with $x = h\frac{c}{\lambda k_B T}$ | wavelength and temperature enter only through x

{chain} 2 $\frac{dB_\lambda}{\mathrm{d}\lambda} = 0$; the chain rule turns the condition into $5(1-e^{-x}) = x$, | peak condition

{chain} 3 the only nonzero root is $x\approx 4.9651$, so $h\frac{c}{\lambda_{\max} k_B T} = 4.9651$ | peak sits at fixed x

{chain} 4 $\lambda_{\max} T = \left(\frac{hc}{4.9651 k_B}\right) = 2.898\times10^{-3}\ \mathrm{m\cdot K}$ | displacement law

Wien-law peak

{chain} 1 $\ln B_\lambda = \ln(2hc^2) - 5\ln\lambda - \left(\frac{hc}{\lambda k_B T}\right)$ | product becomes sum

{chain} 2 $\frac{\mathrm{d}\ln B_\lambda}{\mathrm{d}\lambda} = -\left(\frac{5}{\lambda}\right) + \left(\frac{hc}{\lambda^2 k_B T}\right)$ | slope of log radiance

{chain} 3 $\left(\frac{hc}{\lambda^2 k_B T}\right) = \left(\frac{5}{\lambda}\right)$, one power of λ cancels | peak condition

{chain} 4 $\lambda_{\max} = \left(\frac{hc}{5k_B T}\right)$ | peak wavelength of Wien law

annihilation energy chain

{chain} 1 pair at rest: $2m_e \to 0$, $\Delta m = 2m_e$, $E = \Delta mc^2 = 1.64\times10^{-13}\ \mathrm{J}$ = 1.022 MeV, shared as two 511 keV photons

apply corrections

{chain} 1 lattice fills one octant so divide by 8, two polarisations multiply by 2: $\left(\left(\frac{L}{\pi}\right)\right)^3 4\pi k^2\,\mathrm{d}k\times\left(\frac{2}{8}\right)$ | modes in shell

change variable

{chain} 1 $k = \frac{2\pi}{\lambda}$ with $\left(\frac{\mathrm{d}k}{\mathrm{d}\lambda}\right) = -\left(\frac{2\pi}{\lambda^2}\right)$, so $|\mathrm{d}k| = \left(\frac{2\pi}{\lambda^2}\right)\mathrm{d}\lambda$ | density per unit wavelength

collapse the sums

{chain} 1 $\sum_n z^n = \left(\frac{1}{1-z}\right)$ and $\sum_n nz^n = \left(\frac{z}{(1-z)^2}\right)$ | closed forms for both sums

constants of the integral

{chain} 1 integrating Planck's law gives $\sigma = \left(\frac{2\pi^5 k_B^4}{15h^3c^2}\right)$, reproducing $5.6704\times10^{-8}\frac{\mathrm{W}}{\mathrm{m^2\cdot K^4}}$ | Stefan–Boltzmann constant

convert to radiance

{chain} 1 $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda = \left(\frac{2hc^2}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ | spectral radiance

count oscillator degrees of freedom

{chain} 1 a harmonic oscillator has two, potential and kinetic, so $\left(\frac{1}{2}\right)k_B T + \left(\frac{1}{2}\right)k_B T = k_B T$ | mean mode energy

divide by temperature

{chain} 1 $\mathrm{d}S = \left(\frac{Vu'}{T}\right)\mathrm{d}T + \left(\frac{4u}{3T}\right)\mathrm{d}V$ | entropy differential

divide by volume

{chain} 1 $\left(\frac{1}{L^3}\right)\left(\left(\frac{L}{\pi}\right)\right)^3 4\pi k^2\,\mathrm{d}k\times\left(\frac{2}{8}\right) = \left(\frac{k^2\,\mathrm{d}k}{\pi^2}\right)$ | modes per unit volume

exactness of dS

{chain} 1 $\left(\left(\frac{\partial}{\partial V}\right)\right)_T\left(\left(\frac{\partial S}{\partial T}\right)\right)_V = \left(\left(\frac{\partial}{\partial T}\right)\right)_V\left(\left(\frac{\partial S}{\partial V}\right)\right)_T$. | mixed partials must agree

expand total energy

{chain} 1 $\mathrm{d}U = Vu'(T)\,\mathrm{d}T + u\,\mathrm{d}V$ with $u' = \frac{\mathrm{d}u}{\mathrm{d}T}$ | total differential of U

first law for photon gas

{chain} 1 $\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$ gives $T\,\mathrm{d}S = \mathrm{d}U + p\,\mathrm{d}V$ | entropy relation

geometric substitution

{chain} 1 write $z = e^{-h\frac{f}{k_B T}}$, so the weight of level n is z^n | sums become geometric

group into shell

{chain} 1 modes of one narrow wavelength range occupy shell of radius k and thickness $\mathrm{d}k$, volume $4\pi k^2\,\mathrm{d}k$ | modes by magnitude

hemisphere cosine integral

{chain} 1 flux factor π from $\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\frac{\pi}{2}}\cos\theta\sin\theta\,\mathrm{d}\theta = 2\pi\times\left(\frac{1}{2}\right) = \pi$ | flux from radiance

identify the constant

{chain} 1 $\frac{4hc}{E_0\approx 364.6}\ \mathrm{nm}$ matches Balmer's fitted 364.56 nm | fitted constant from constants

insert into the shared chain

{chain} 1 $I_\lambda = \left(\frac{2\pi c}{\lambda^4}\right)\overline{E}$ with $\overline{E} = k_B T$ gives $I_\lambda = \left(\frac{2\pi c k_B T}{\lambda^4}\right)$ | Rayleigh–Jeans law

integrate

{chain} 1 $\frac{\mathrm{d}u}{u} = 4\frac{\mathrm{d}T}{T}$ gives $u = aT^4$ with a a constant, and $R = \sigma T^4$ | temperature dependence of energy density

integrate the spectrum

{chain} 1 $\int_0^\infty I_\lambda(\lambda, T)\,\mathrm{d}\lambda$ over every wavelength inherits T^4 from the curve | total power scaling

lattice density

{chain} 1 permitted points form cubic lattice of spacing $\frac{\pi}{L}$, filling k-space at $\left(\frac{L}{\pi}\right)^3$ points per unit volume | uniform k-space filling

level drop

{chain} 1 a drop from level k to the second level releases $\frac{hc}{\lambda} = E_0\left(\frac{1}{4} - \frac{1}{k^2}\right)$ | photon energy

long-wavelength limit

{chain} 1 $e^{h\frac{f}{k_B T}} - 1 \approx h\frac{f}{k_B T}$ makes $E(f) \to k_B T$, giving $I_\lambda = \left(\frac{2\pi c k_B T}{\lambda^4}\right)$ | Rayleigh–Jeans law recovered

multiply by mode density

{chain} 1 $u_\lambda = \left(\frac{8\pi hc}{\lambda^5}\right)\left(\frac{1}{e^{h\frac{c}{\lambda k_B T}}-1}\right)$ with $f = \frac{c}{\lambda}$ | energy density

pair-production momentum chain

{chain} 1 photon carries $\frac{E}{c}$; a third body (nucleus) must absorb recoil, else pair cannot form in empty space (2 MeV photon in vacuum produces nothing)

pair-production threshold chain

{chain} 1 photon must carry at least combined rest energy $2m_e c^2 = 1.022$ MeV; below it no event

photon wavelength chain

{chain} 1 $E = \frac{hc}{\lambda}$ rearranges to $\lambda = \frac{hc}{E} = (6.626 \times 10^{-34})\frac{3\times10^8}{8.2\times10^{-14}} = 2.4 \times 10^{-12}$ m

photon-gas pressure

{chain} 1 rays at angle θ carry normal momentum scaling as $\cos^2\theta$; hemisphere average is $\frac{1}{3}$, so $p = \frac{u}{3}$ | radiation pressure

produce the fourth power

{chain} 1 $\left(\frac{k^2\,\mathrm{d}k}{\pi^2}\right) = \left(\frac{4\pi^2}{\lambda^2}\right) \times \left(\frac{2\pi}{\lambda^2}\right)\mathrm{d}\lambda \times \left(\frac{1}{\pi^2}\right) = \left(\frac{8\pi}{\lambda^4}\right)\mathrm{d}\lambda \,|\, n(\lambda) = \frac{8\pi}{\lambda^4}$

read partial derivatives

{chain} 1 $\left(\left(\frac{\partial S}{\partial T}\right)\right)_V = V\frac{u'}{T}$ and $\left(\left(\frac{\partial S}{\partial V}\right)\right)_T = 4\frac{u}{3T}$ | entropy partials

rearrange for wavelength

{chain} 1 $\lambda = \left(\frac{4hc}{E_0}\right)\frac{k^2}{k^2-4}$ | Balmer formula shape

rewrite validity on wavelength

{chain} 1 $f = \frac{c}{\lambda}$ turns $hf \ll k_B T$ into $\lambda \gg h\frac{c}{k_B T} = 1.4388 \times 10^{-2}$ m $\cdot \frac{\mathrm{K}}{T}$ | where the law holds

short-wavelength limit

{chain} 1 the 1 in the denominator becomes negligible, law falls off as $e^{-h\frac{c}{\lambda k_B T}}$ | Wien form recovered

solid-angle conversion

{chain} 1 $u_\lambda = n(\lambda)\overline{E}$, $B_\lambda = \left(\frac{c}{4\pi}\right)u_\lambda$ from cavity isotropy over 4π steradians | energy density and radiance

standing-wave condition

{chain} 1 field vanishes at opposite walls, so $k_x = \left(\frac{\pi n_x}{L}\right)$ with positive integers n_x, n_y, n_z | permitted wavevectors

substitute energy and pressure

{chain} 1 $T\,\mathrm{d}S = Vu'\,\mathrm{d}T + \left(\frac{4}{3}\right)u\,\mathrm{d}V.$ | first law for photon gas

substitute the partials

{chain} 1 $\left(\frac{u'}{T}\right) = \left(\frac{4}{3}\right)\left(\left(\frac{u'}{T}\right) - \left(\frac{u}{T^2}\right)\right)$, which rearranges to $Tu' = 4u$ | differential equation for u

take the ratio

{chain} 1 $E(f) = hf\left(\frac{\frac{z}{(1-z)^2}}{\frac{1}{1-z}}\right) = \left(\frac{hfz}{1-z}\right) = \left(\frac{hf}{e^{h\frac{f}{k_B T}}-1}\right)$ | mean energy closed form

Week 3 lecture 1

the planetary atom's failure; Geiger–Marsden backscattering; Rutherford's scattering equation and cross sections

Rutherford scattering experiments

A219 $K = \left(\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 a}\right), \quad a = \left(\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 K}\right)$ head-on distance of closest approach *(Rutherford scattering experiments)*

A220 $E = K = \left(\frac{1}{2}\right)mv_0^2$, $L = mv_0 b$ conserved energy and angular momentum *(Rutherford scattering experiments)*

A221 $k_C = \left(\frac{Z_1 Z_2 e^2}{4\pi\epsilon_0}\right)$ repulsive Coulomb constant *(Rutherford scattering experiments)*

A222 $\left(\frac{1}{2}\right)m\left(\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)\right)^2 + \left(\frac{L^2}{2mr^2}\right) + \left(\frac{k_C}{r}\right) = E$ radial energy equation *(Rutherford scattering experiments)*

A223 $\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right) = -\left(\frac{L}{m}\right)\left(\frac{\mathrm{d}u}{\mathrm{d}\phi}\right)$, $u = \frac{1}{r}$ radial speed in orbit variable *(Rutherford scattering experiments)*

A224 $\left(\left(\frac{\mathrm{d}u}{\mathrm{d}\phi}\right)\right)^2 + u^2 + \left(\frac{2mk_C}{L^2}\right)u = \left(\frac{2mE}{L^2}\right)$ energy equation in u *(Rutherford scattering experiments)*

A225 $\left(\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}\right) + u = -\left(\frac{mk_C}{L^2}\right)$ orbit equation *(Rutherford scattering experiments)*

A226 $u = \left(\frac{mk_C}{L^2}\right)(e\cos\phi - 1)$, $e = \sqrt{1 + \left(\frac{2EL^2}{mk_C^2}\right)}$ hyperbolic orbit and eccentricity *(Rutherford scattering experiments)*

A227 $\theta = \pi - 2\phi_\infty$, $\tan\left(\frac{\theta}{2}\right) = \cot\phi_\infty = \left(\frac{1}{\sqrt{e^2-1}}\right)$ asymptote angle and deflection *(Rutherford scattering experiments)*

A228 $e^2 - 1 = \left(\frac{m^2 v_0^4 b^2}{k_C^2}\right) = \left(\frac{4b^2}{a^2}\right)$ eccentricity in terms of impact parameter *(Rutherford scattering experiments)*

A229 $b = \left(\frac{a}{2}\right)\cot\left(\frac{\theta}{2}\right) = \left(\frac{Z_1 Z_2 e^2}{8\pi\epsilon_0 K}\right)\cot\left(\frac{\theta}{2}\right)$ impact parameter locked to scattering angle *(Rutherford scattering experiments)*

A230 $b = 0$ gives $\theta = 180°$ head-on particle returns straight back *(Rutherford scattering experiments)*

A231 $\sigma = \pi b^2$ cross section for scattering at angle θ or more *(Rutherford scattering experiments)*

A232 $f = nt\sigma = nt\pi b^2$ fraction of beam scattered beyond θ *(Rutherford scattering experiments)*

A233 $\mathrm{d}\sigma = 2\pi b \,|\,\mathrm{d}b|$ annulus cross-section element *(Rutherford scattering experiments)*

A234 $\left(\frac{\mathrm{d}b}{\mathrm{d}\theta}\right) = -\left(\frac{a}{4}\right)\csc^2\left(\frac{\theta}{2}\right)$ impact-parameter derivative *(Rutherford scattering experiments)*

A235 $\mathrm{d}\Omega = 2\pi\sin\theta\,\mathrm{d}\theta$ solid-angle ring *(Rutherford scattering experiments)*

A236 $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right) = \left(\frac{b}{\sin\theta}\right) \,|\, \left(\frac{\mathrm{d}b}{\mathrm{d}\theta}\right) \,| = \left(\left(\frac{a}{4}\right)\right)^2\left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)}\right)$ differential cross section *(Rutherford scattering experiments)*

A237 $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right) = \left(\left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 K}\right)\right)^2\left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)}\right)$ Rutherford scattering equation *(Rutherford scattering experiments)*

A238 detector count $N_i nt\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)\mathrm{d}\Omega$ arrivals from N_i incident particles *(Rutherford scattering experiments)*

A239 about 1 in 8,000 alphas bounce straight back 1909 backscattering count *(Rutherford scattering experiments)*

A240 $\theta_{\mathrm{rms}} = \delta\sqrt{N}$ random-walk quadrature total *(Rutherford scattering experiments)*

A241 per gold atom $0.016° \times \sqrt{79} \approx 0.14°$ plum-pudding per-atom deflection *(Rutherford scattering experiments)*

A242 $0.14° \times \sqrt{2\{,\}300} \approx 6.7° \approx 7°$ film random-walk total, too small *(Rutherford scattering experiments)*

A243 single electron collision $0.016°$; 79 electrons $\approx 0.14°$ plum-pudding deflection scales *(Rutherford scattering experiments)*

A244 $Z_1 = 4$ (alpha), $Z_2 = 79$ (gold), K near 7.7 MeV, $t = 0.6\mu$ m test-condition numbers *(Rutherford scattering experiments)*

A245 $K_{\mathrm{kinematic}} = \frac{E_{\mathrm{out}}}{E_{\mathrm{in}}}$ RBS kinematic factor, heavier target gives larger K *(Rutherford scattering experiments)*

Geiger-Marsden alpha scattering (Rutherford scattering experiments): fire alpha beam at thin gold foil, count particles by direction
Measure: fraction backscattered (1 in 8000) and angular distribution. Determines: nuclear atom: point-like nucleus smaller than distance a.

Rutherford backscattering spectrometry (Rutherford scattering experiments): aim alpha beam into sample, count backscattered particles and their energies
Measure: count (how much of what) and energy (which element, how deep). Determines: composition and depth profile, non-destructive, count ∝ Z_2^2.

Derivation chains

Rutherford b(θ) hyperbola chain

{chain} 1 off-centre orbit conserves $E = \left(\frac{1}{2}\right)mv_0^2$ and $L = mv_0 b$; charges collected into $k_C = Z_1 Z_2 \frac{e^2}{4\pi\epsilon_0}$

{chain} 2 $\left(\frac{1}{2}\right)m\left(\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)\right)^2 + \left(\frac{L^2}{2mr^2}\right) + \left(\frac{k_C}{r}\right) = E$

{chain} 3 $\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right) = \left(\frac{\mathrm{d}\left(\frac{1}{u}\right)}{\mathrm{d}\phi}\right)\left(\frac{\mathrm{d}\phi}{\mathrm{d}t}\right) = -\left(\frac{1}{u^2}\right)\left(\frac{\mathrm{d}u}{\mathrm{d}\phi}\right) \cdot \left(\frac{L}{mr^2}\right) = -\left(\frac{L}{m}\right)\left(\frac{\mathrm{d}u}{\mathrm{d}\phi}\right)$, since $\frac{1}{u^2} = r^2$ cancels the $\frac{1}{r^2}$

{chain} 4 substituting gives $\left(\left(\frac{\mathrm{d}u}{\mathrm{d}\phi}\right)\right)^2 + u^2 + \left(\frac{2mk_C}{L^2}\right)u = \left(\frac{2mE}{L^2}\right)$

{chain} 5 term-by-term differentiation and division by $2\frac{\mathrm{d}u}{\mathrm{d}\phi}$ (RHS constant) leaves $\left(\frac{d^2u}{\mathrm{d}\phi^2}\right) + u = -\left(\frac{mk_C}{L^2}\right)$

{chain} 6 a constant has zero second derivative, so the equation itself fixes $u_p = -\left(\frac{mk_C}{L^2}\right)$; homogeneous part $R\cos(\phi - \phi_0)$

{chain} 7 $\phi = 0$ at closest approach sets $\phi_0 = 0$ ($\frac{\mathrm{d}u}{\mathrm{d}\phi} = 0$); feeding back fixes $R = \frac{emk_C}{L^2}$, so $u = \left(\frac{mk_C}{L^2}\right)(e\cos\phi - 1)$ with $e = \sqrt{1 + \left(\frac{2EL^2}{mk_C^2}\right)} > 1$

{chain} 8 $u = 0$ gives $\cos\phi_\infty = \frac{1}{e}$; $\theta = \pi - 2\phi_\infty$ and $\cot\phi_\infty = \left(\frac{\frac{1}{e}}{\sqrt{1-\frac{1}{e^2}}}\right) = \left(\frac{1}{\sqrt{e^2-1}}\right)$

{chain} 9 $L = mv_0 b$, $E = \left(\frac{1}{2}\right)mv_0^2$ give $e^2 - 1 = \left(\frac{m^2 v_0^4 b^2}{k_C^2}\right) = \left(\frac{4b^2}{a^2}\right)$ with $a = \frac{k_C}{E}$, so $\tan\left(\frac{\theta}{2}\right) = \left(\frac{a}{2b}\right)$, i.e. $b = \left(\frac{a}{2}\right)\cot\left(\frac{\theta}{2}\right)$

dσ/dΩ chain

{chain} 1 impact parameters b to $b + \mathrm{d}b$ form annulus of area $2\pi b\,\mathrm{d}b$; since b falls as θ grows, $\mathrm{d}\sigma = 2\pi b \,|\,\mathrm{d}b|$

{chain} 2 ring of directions covers $\mathrm{d}\Omega = 2\pi\sin\theta\,\mathrm{d}\theta$, with $\sin\theta = 2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$

{chain} 3 $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right) = \left(\frac{2\pi b\,|\,\mathrm{d}b|}{2\pi\sin\theta\,\mathrm{d}\theta}\right) = \left(\frac{b}{\sin\theta}\right) \,|\, \left(\frac{\mathrm{d}b}{\mathrm{d}\theta}\right) \,|$, the 2π cancels

{chain} 4 with $b = \left(\frac{a}{2}\right)\cot\left(\frac{\theta}{2}\right)$ and $\left|\,\left(\frac{\mathrm{d}b}{\mathrm{d}\theta}\right)\,\right| = \left(\frac{a}{4}\right)\csc^2\left(\frac{\theta}{2}\right)$, the $\cos\left(\frac{\theta}{2}\right)$ of the cotangent cancels the one inside $\sin\theta$, leaving $\left(\left(\frac{a}{4}\right)\right)^2\left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)}\right)$

{chain} 5 $\frac{a}{4} = \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 K}\right)$ gives $\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right) = \left(\left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 K}\right)\right)^2\left(\frac{1}{\sin^4\left(\frac{\theta}{2}\right)}\right)$; counts follow as $N_i nt\left(\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right)\mathrm{d}\Omega$

random-walk quadrature arithmetic

{chain} 1 N uncorrelated kicks of size δ add as $\theta_{\mathrm{rms}} = \delta\sqrt{N}$ (linear sum $N\delta$ overcounts): $0.016° \times \sqrt{79} \approx 0.14°$ per atom; $0.14° \times \sqrt{2300} \approx 0.14° \times 48 \approx 6.7° \approx 7°$ through the film, far below backscatter

Week 3 lecture 2

Bohr's hydrogen model; the fine-structure constant; Franck–Hertz; Moseley's law; the correspondence principle

Bohr model

A246 $\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$ Coulomb attraction supplies centripetal force *(Bohr model)*

A247 $K = \frac{e^2}{8\pi\epsilon_0 r}$ planetary electron kinetic energy *(Bohr model)*

A248 $E = -\frac{e^2}{8\pi\epsilon_0 r}$ planetary electron total energy *(Bohr model)*

A249 $hf = E_u - E_l$ photon energy on a level transition *(Bohr model)*

A250 $mvr = n\hbar$ angular-momentum quantisation condition *(Bohr model)*

A251 $r_n = 4\pi\epsilon_0 n^2\frac{\hbar^2}{me^2} = n^2 a_0$ allowed Bohr orbit radii *(Bohr model)*

A252 $a_0 = \left(\frac{4\pi\epsilon_0\hbar^2}{me^2}\right) = 0.53$ Å Bohr radius numerical value *(Bohr model)*

A253 $v_n = \frac{e^2}{4\pi\epsilon_0\hbar n}$ electron speed falls as 1/n *(Bohr model)*

A254 $\alpha = \left(\frac{v_1}{c}\right) = \left(\frac{e^2}{4\pi\epsilon_0\hbar c}\right) \approx \left(\frac{1}{137}\right)$ fine structure constant *(Bohr model)*

A255 $E_n = -\left(\frac{e^2}{8\pi\epsilon_0 a_0 n^2}\right) = \left(\frac{E_1}{n^2}\right)$ stationary-state energy ladder *(Bohr model)*

A256 $E_1 = -\frac{e^2}{8\pi\epsilon_0 a_0} = -13.6$ eV ground-state energy of hydrogen *(Bohr model)*

A257 $\left(\frac{1}{\lambda}\right) = R_\infty\left(\left(\frac{1}{n_l^2}\right) - \left(\frac{1}{n_u^2}\right)\right)$ spectral series wavenumber formula *(Bohr model)*

A258 $R_\infty \approx 1.097 \times 10^7 \mathrm{m}^{-1}$ Rydberg constant value *(Bohr model)*

A259 $r = 0.5$ Å gives $v \approx 2.2 \times 10^6$ m/s, $E \approx -14$ eV planetary electron numbers in hydrogen *(Bohr model)*

A260 classical lifetime $\approx 10^{-11}$ s atom collapse estimate from radiation *(Bohr model)*

A261 $r_n = \frac{n^2 a_0}{Z}$ and $E_n = -\frac{13.6Z^2}{n^2}$ eV hydrogen-like ion extension *(Bohr model)*

Franck–Hertz experiment

A262 drops at 4.9 V, 9.8 V, 14.7 V current-drop voltages *(Franck–Hertz experiment)*

A263 $\Delta E = 4.9$ eV energy increment surrendered each drop *(Franck–Hertz experiment)*

A264 mercury first excited state 4.9 eV above ground excitation energy deciding collisions *(Franck–Hertz experiment)*

A265 $\lambda = 254$ nm ultraviolet line excited-mercury emission wavelength *(Franck–Hertz experiment)*

A266 $E = 1240\frac{\text{eV nm}}{254}$ nm ≈ 4.9 eV photon energy matching the step *(Franck–Hertz experiment)*

Franck-Hertz (Franck–Hertz experiment): fire electrons through mercury vapor, raise accelerating voltage, record collector current plus emitted light
Measure: current vs voltage drops and UV wavelength. Determines: discrete excitation energy 4.9 eV of mercury.

Derivation chains

Bohr radius and energy ladder

{chain} 1 Newton plus Coulomb: $\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$ gives $K = \frac{e^2}{8\pi\epsilon_0 r}$ and $E = K + V = -\frac{e^2}{8\pi\epsilon_0 r}$

{chain} 2 $mvr = n\hbar$ gives $v = n\frac{\hbar}{mr}$

{chain} 3 $v = n\frac{\hbar}{mr}$ into force balance gives $n^2\frac{\hbar^2}{mr^3} = \frac{e^2}{4\pi\epsilon_0 r^2}$

{chain} 4 allowed radii $r_n = 4\pi\epsilon_0 n^2\frac{\hbar^2}{me^2} = n^2 a_0$ with $a_0 = 4\pi\epsilon_0\frac{\hbar^2}{me^2}$

{chain} 5 force balance with allowed radii gives $v_n = \frac{e^2}{4\pi\epsilon_0\hbar n}$; $v_1 = 2.2 \times 10^6$ m/s

{chain} 6 insert r_n into $E = -\frac{e^2}{8\pi\epsilon_0 r}$ gives $E_n = \frac{E_1}{n^2}$, $E_1 = -13.6$ eV

{chain} 7 $hf = E_u - E_l$ with E_n gives $\frac{1}{\lambda} = R_\infty\left(\frac{1}{n_l^2} - \frac{1}{n_u^2}\right)$, $R_\infty = \frac{E_0}{hc}$, $E_0 = 13.6$ eV

correspondence-principle limit

{chain} 1 $E_{n+1} - E_n = E_1\left[\frac{1}{(n+1)^2} - \frac{1}{n^2}\right] = -E_1\frac{2n+1}{n^2(n+1)^2}$

{chain} 2 $\frac{2n+1}{n^2(n+1)^2}$ behaves as $\frac{2}{n^3}$, and $E_1 < 0$ makes $-E_1 > 0$, so $E_{n+1} - E_n$ falls as $2|E_1|\frac{1}{n^3}$

{chain} 3 $f = |E_{n+1} - E_n|\frac{1}{h} \approx 2|E_1|\frac{1}{\hbar n^3}$ for a drop from $n+1$

{chain} 4 $\omega = \sqrt{\frac{e^2}{4\pi\epsilon_0 mr^3}}$ with $r = n^2 a_0$; r^3 grows as n^6 so $f_{\text{orb}} = \frac{\omega}{2\pi}$ falls as $\frac{1}{n^3}$ with matching coefficient

Franck-Hertz light-energy agreement

{chain} 1 current steps give ΔE = 4.9 eV; emitted line at 254 nm has $E = \frac{hc}{\lambda} = 1240\frac{\text{eV nm}}{254}$ nm ≈ 4.9 eV, one number measured two ways

Week 4 lecture 1

X-ray diffraction and Bragg's law; the de Broglie hypothesis and electron diffraction; standing-wave quantisation

Bragg's law

A267 $n\lambda = 2d\sin\theta$ constructive reflection condition from lattice planes *(Bragg's law)*

A268 path difference $2d\sin\theta$ extra path of deeper-plane ray *(Bragg's law)*

A269 Wien/Sommerfeld estimate 10^{-10} to 10^{-11} m X-ray wavelength scale *(Bragg's law)*

A270 $d_{hkl} = \left(\frac{a}{\sqrt{h^2+k^2+l^2}}\right)$ cubic plane-family spacing *(Bragg's law)*

A271 (200): $\frac{a}{2}$; (220): $\frac{a}{\sqrt{8}}$; (222): $\frac{a}{\sqrt{12}}$; (400): $\frac{a}{4}$; (111): $\frac{a}{\sqrt{3}}$ NaCl-family spacings *(Bragg's law)*

A272 Cu $K\alpha\lambda = 1.54$ Å, NaCl peak $2\theta = 31.7°$ (200) powder pattern reference numbers *(Bragg's law)*

A273 $d_{200} = \frac{\lambda}{2\sin\theta} = 2.82$ Å, $a = 2d_{200} = 5.64$ Å NaCl lattice-parameter arithmetic *(Bragg's law)*

A274 $a = d_{hkl}\sqrt{h^2+k^2+l^2}$ peak-to-family consistency check *(Bragg's law)*

A275 $d_{2h,2k,2l} = \frac{d_{hkl}}{2}$ doubled indices halve spacing *(Bragg's law)*

Bragg reflection (Bragg's law): rotate sample and detector, record diffracted intensity against scattering angle
Measure: peak positions where $n\lambda = 2d\sin\theta$ holds. Determines: d-spacings, hence lattice parameters.

Bragg transmission (Bragg's law): pass beam through powder onto flat plate behind sample
Measure: concentric ring radii around undeflected spot. Determines: d-spacings, ring radius growing with 2θ.

matter wave

A276 $\lambda = \frac{h}{p}$, $\lambda = \frac{h}{mv}$ de Broglie wavelength *(matter wave)*

A277 $h = 6.626 \times 10^{-34}$ J s Planck constant value *(matter wave)*

A278 $E = hf$ and $E = pc$ give $p = \frac{hf}{c}$, then $\lambda = \frac{h}{p}$ photon route to de Broglie relation *(matter wave)*

A279 runner 100 kg at 10 m/s: $\lambda \approx 7 \times 10^{-37}$ m everyday-object wavelength *(matter wave)*

A280 electron 10^{-30} kg at 5×10^6 m/s: $\lambda \approx 1.3 \times 10^{-10}$ m atomic-scale electron wavelength *(matter wave)*

A281 $n\lambda = D\sin\phi$ surface grating diffraction condition *(matter wave)*

A282 $\phi = 50°$, $D = 2.15$ Å give $\lambda \approx 0.165$ nm Davisson-Germer measured wavelength *(matter wave)*

A283 $\left(\frac{1}{2}\right)mv^2 = eV$, $v = \sqrt{\frac{2eV}{m}}$ electron accelerated through voltage V *(matter wave)*

A284 $p = mv = \sqrt{2meV}$ accelerated-electron momentum *(matter wave)*

A285 $\lambda = \frac{h}{\sqrt{2meV}}$ accelerated-electron de Broglie wavelength *(matter wave)*

A286 $V = 55$ V gives $\lambda \approx 0.165$ nm Davisson-Germer agreement number *(matter wave)*

A287 $V = 150$ V gives $\lambda \approx 0.1$ nm $= 1$ Å G. P. Thomson ring-pattern match *(matter wave)*

A288 $2\pi r = n\lambda$ circular standing-wave resonance condition *(matter wave)*

A289 $L = rp = \frac{nh}{2\pi} = n\hbar$ Bohr quantisation derived from resonance *(matter wave)*

A290 $\lambda_n = \frac{2\pi r_n}{n} = 2\pi n a_0$; first three $6.28a_0$, $12.57a_0$, $18.85a_0$ hydrogen orbit wavelengths *(matter wave)*

Davisson-Germer (matter wave): fire electron beam at nickel single crystal, scan scattering angle
Measure: scattered intensity vs angle. Determines: electron wavelength via $n\lambda = D\sin\phi$, tested against $\lambda = \frac{h}{\sqrt{2meV}}$.

G. P. Thomson diffraction (matter wave): pass electrons through polycrystalline thin metal foil
Measure: photograph diffraction rings. Determines: electron wavelength (rings match X-rays of same λ).

LEED (matter wave): aim slow electrons at crystal surface, record diffracted beams
Measure: spot positions. Determines: surface periodicity via $n\lambda = D\sin\phi$; spot motion tracks voltage.

Derivation chains

p = √(2meV) chain

{chain} 1 electron from rest through V gains eV, so $\left(\frac{1}{2}\right)mv^2 = eV$

{chain} 2 $v = \sqrt{\frac{2eV}{m}}$

{chain} 3 $p = mv = \sqrt{2meV}$

{chain} 4 $\lambda = \frac{h}{\sqrt{2meV}}$; at 55 V ≈ 0.165 nm, matching the grating value

Bragg path difference

{chain} 1 deeper ray gains $d\sin\theta$ before reflection and $d\sin\theta$ after, so reflections add only when $2d\sin\theta = n\lambda$

NaCl lattice parameter

{chain} 1 $\theta = 15.85°$ gives $d_{200} = \frac{1.54}{2\sin 15.85°} = 2.82$ Å; cubic (200) spacing is $\frac{a}{2}$, so $a = 5.64$ Å

circular standing-wave chain

{chain} 1 $2\pi r = n\lambda$ with $\lambda = \frac{h}{p}$ gives $2\pi r = \frac{nh}{p}$, so $L = rp = n\hbar$; with $r_n = n^2 a_0$, $\lambda_n = 2\pi n a_0$

photon derivation of λ = h/p

{chain} 1 $E = hf$ and $E = pc$ give $p = \frac{hf}{c}$; with $\lambda = \frac{c}{f}$ eliminating f and c leaves $\lambda = \frac{h}{p}$

Week 4 lecture 2

the wave function and Born's rule; uncertainty; Copenhagen; wave packets; the particle in a box

wave function

A291 $|\Psi(x,t)|^2$ probability density at position x and time t *(wave function)*

A292 $\int_a^b |\Psi(x,t)|^2\,\mathrm{d}x$ probability of finding particle between a and b *(wave function)*

A293 $\int_{-\infty}^{\infty} |\Psi(x,t)|^2\,\mathrm{d}x = 1$ normalisation condition *(wave function)*

A294 $\langle x\rangle = \int x\,|\Psi(x,t)|^2\,\mathrm{d}x$ mean position *(wave function)*

A295 $\langle g(x)\rangle = \int g(x)\,|\Psi(x,t)|^2\,\mathrm{d}x$ mean of function of position *(wave function)*

A296 $\Delta x = \sqrt{\langle x^2\rangle - \langle x\rangle^2}$ position spread from moments *(wave function)*

A297 $\hat{p} = -i\hbar\frac{\partial}{\partial x}$ operator needed for momentum averages *(wave function)*

A298 $|\Psi|^2 = |\Psi_1|^2 + |\Psi_2|^2 + 2\,\mathrm{Re}\,(\Psi_1\Psi_2^*)$ superposition interference decomposition *(wave function)*

uncertainty principle

A299 $\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$ position-momentum uncertainty bound *(uncertainty principle)*

A300 $\hbar = \frac{h}{2\pi}$ reduced Planck constant *(uncertainty principle)*

A301 $\Psi(x) = Ae^{-\frac{x^2}{4}(\Delta x)^2}e^{ik_0 x}$ Gaussian packet saturating Fourier bound *(uncertainty principle)*

A302 $|\Psi|^2 \propto e^{-\frac{x^2}{2}(\Delta x)^2}$ Gaussian density with standard deviation Δx *(uncertainty principle)*

A303 $\Delta k = \frac{1}{2\Delta x}$ and $\Delta k \cdot \Delta x = \frac{1}{2}$ exactly Gaussian saturation of Fourier bound *(uncertainty principle)*

A304 $\Delta p_x = \hbar\Delta k$ momentum spread from wave-number spread *(uncertainty principle)*

A305 $\frac{\theta \approx \lambda}{a}$ single-slit diffraction angle *(uncertainty principle)*

A306 $\frac{\Delta y \approx \lambda L}{a}$ single-slit band width at distance L *(uncertainty principle)*

A307 $\frac{\Delta p_y \approx h}{a}$ transverse momentum spread from slit width *(uncertainty principle)*

A308 $\Delta y \cdot \Delta p_y \approx h$ slit estimate matching uncertainty bound *(uncertainty principle)*

A309 $\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$ energy-time uncertainty relation *(uncertainty principle)*

A310 $\Delta E \geq \frac{\hbar}{2\Delta t}$ natural linewidth from finite state lifetime *(uncertainty principle)*

A311 $\Delta x \cdot \Delta p_x \gtrsim h$ gamma-ray microscope estimate *(uncertainty principle)*

A312 photon momentum $\frac{h}{\Delta x}$ microscope kick lower bound *(uncertainty principle)*

A313 $\Delta x \approx a_0 \approx 0.53 \times 10^{-10}$ m, $\frac{\Delta p_x \approx \hbar}{a_0 \approx 10^{-24}}$ kg·m/s, $v \approx 2 \times 10^{6}$ m/s atomic-scale confinement numbers *(uncertainty principle)*

Copenhagen interpretation

A314 $\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$ uncertainty bound as pillar one *(Copenhagen interpretation)*

A315 Born rule: $|\Psi|^2$ probability measure pillar three on outcomes *(Copenhagen interpretation)*

A316 complementarity: diffraction setup shows no path, which-path setup no interference pillar two in experimental form *(Copenhagen interpretation)*

wave packet

A317 $\Psi(x,0) = Ae^{-\frac{x^2}{4}(\Delta x)^2} e^{ik_0 x}$ Gaussian packet at time zero *(wave packet)*

A318 $\Delta k \sim \frac{1}{2\Delta x}$ Fourier spread of Gaussian envelope *(wave packet)*

A319 $\Delta k \cdot \Delta x \geq \frac{1}{2}$ Fourier localisation bound *(wave packet)*

A320 $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$ uncertainty relation from Fourier bound *(wave packet)*

A321 $u = \frac{\Delta\omega}{\Delta k} \to \frac{\mathrm{d}\omega}{\mathrm{d}k}$ group velocity from two-wave beats *(wave packet)*

A322 $v_{\mathrm{ph}} = \frac{\omega}{k}$ phase velocity of carrier crests *(wave packet)*

A323 $\omega = \alpha k$ gives $u = v_{\mathrm{ph}} = \alpha$ non-dispersive case *(wave packet)*

A324 $\omega = \beta k^2$ gives $u = 2v_{\mathrm{ph}}$ quadratic dispersion case *(wave packet)*

A325 $\omega = \gamma\sqrt{k}$ gives $u = \frac{v_{\mathrm{ph}}}{2}$ square-root dispersion case *(wave packet)*

A326 $\omega = \frac{\hbar k^2}{2m}$ free non-relativistic matter-wave dispersion *(wave packet)*

A327 $u = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{\hbar k}{m} = \frac{p}{m}$ matter-wave group velocity equals classical speed *(wave packet)*

A328 $v_{\mathrm{ph}} = \frac{\omega}{k} = \frac{\hbar k}{2m}$ matter-wave phase velocity, half of u *(wave packet)*

particle in a box

A329 $E_n = n^2 \frac{h^2}{8mL^2}$ infinite-well energy levels *(particle in a box)*

A330 $E_1 = \frac{h^2}{8mL^2}$ nonzero ground-state energy *(particle in a box)*

A331 $n\left(\frac{\lambda}{2}\right) = L$, $\lambda_n = 2\frac{L}{n}$ standing-wave quantisation condition *(particle in a box)*

A332 $p_n = n\frac{h}{2L}$ de Broglie momentum of box standing wave *(particle in a box)*

A333 $-\left(\frac{\hbar^2}{2m}\right)\left(\frac{d^2\psi}{dx^2}\right) = E\psi$ Schrödinger equation inside the well *(particle in a box)*

A334 $k = \frac{\sqrt{2mE}}{\hbar}$ wave number inside the well *(particle in a box)*

A335 $k_n = \frac{n\pi}{L}$ boundary-condition wave numbers *(particle in a box)*

A336 $E_n = n^2\pi^2\frac{\hbar^2}{2mL^2} = n^2\frac{h^2}{8mL^2}$ levels from k_n *(particle in a box)*

A337 $\int_0^L \sin^2\left(\frac{n\pi x}{L}\right)\mathrm{d}x = \frac{L}{2}$ normalisation integral *(particle in a box)*

A338 $\psi_{n(x)} = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$ normalised box eigenfunctions *(particle in a box)*

A339 $|\psi_{n(x)}|^2 = \left(\frac{2}{L}\right)\sin^2\left(\frac{n\pi x}{L}\right)$, $n-1$ interior nodes level densities and node count *(particle in a box)*

A340 $\langle x\rangle = \frac{L}{2}$, $\langle p\rangle = 0$, $\langle p^2\rangle = 2mE_n$ box level averages *(particle in a box)*

A341 string: $f_n = \frac{nv}{2}L$, same $\lambda_n = 2\frac{L}{n}$ vibrating-string correspondence *(particle in a box)*

A342 $E = E_x + E_y + E_z$ with $E_i = n_i^2\pi^2\frac{\hbar^2}{2mL_i^2}$ rectangular 3D box *(particle in a box)*

A343 $E = \left(\frac{\pi^2\hbar^2}{2mL^2}\right)\left(n_x^2 + n_y^2 + n_z^2\right)$, $\psi = A\sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right)$ cubic box spectrum *(particle in a box)*

A344 $(2,1,1)$, $(1,2,1)$, $(1,1,2)$ share $E = \left(\frac{6\pi^2\hbar^2}{2mL^2}\right)$ cubic degeneracy at sum of squares 6 *(particle in a box)*

Derivation chains

box Schrödinger derivation to E_n

{chain} 1 inside well: $-\left(\frac{\hbar^2}{2m}\right)\psi'' = E\psi$, i.e. $\psi'' = -k^2\psi$ with $k = \frac{\sqrt{2mE}}{\hbar}$

{chain} 2 $\psi = Ae^{ikx} + Be^{-ikx}$; Euler $e^{\pm ikx} = \cos kx \pm i\sin kx$ shows $\psi = A\sin kx + B\cos kx$ spans the same space

{chain} 3 $\psi(0) = 0$ kills the cosine: $B = 0$

{chain} 4 $\psi(L) = 0$ forces $\sin kL = 0$, so $k_n = \frac{n\pi}{L}$, $n = 1, 2, 3, \ldots$

{chain} 5 k_n into $E = \hbar^2\frac{k^2}{2}m$ gives $E_n = n^2\pi^2\frac{\hbar^2}{2mL^2} = n^2\frac{h^2}{8mL^2}$

{chain} 6 $\int_0^L \sin^2\left(\frac{n\pi x}{L}\right)\mathrm{d}x = \frac{L}{2}$ gives $\psi_{n(x)} = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$; time dependence via $e^{-\frac{iE_n t}{\hbar}}$

uncertainty Gaussian widths

{chain} 1 $\Psi = Ae^{-\frac{x^2}{4}(\Delta x)^2}e^{ik_0 x}$ gives $|\Psi|^2 \propto e^{-\frac{x^2}{2}(\Delta x)^2}$, standard deviation Δx

{chain} 2 envelope Fourier transform is Gaussian in k with $\Delta k = \frac{1}{2\Delta x}$

{chain} 3 $\Delta k \cdot \Delta x = \frac{1}{2}$ exactly; every other shape gives more

{chain} 4 $p = \hbar k$ gives $\Delta p_x = \hbar\Delta k$, so $\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$

beats group velocity

{chain} 1 superpose $\omega_0 \pm \Delta\omega$, $k_0 \pm \Delta k$; envelope speed $u = \frac{\Delta\omega}{\Delta k}$ tends to $\frac{\mathrm{d}\omega}{\mathrm{d}k}$

matter-wave dispersion

{chain} 1 $E = \frac{p^2}{2}m$ with $E = \hbar\omega$ and $p = \hbar k$ gives $\omega = \frac{\hbar k^2}{2}m$

{chain} 2 $u = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{\hbar k}{m} = \frac{p}{m}$

{chain} 3 $v_{\mathrm{ph}} = \frac{\omega}{k} = \frac{\hbar k}{2}m$, half the group velocity

single-slit chain

{chain} 1 slit width a gives $\frac{\theta \approx \lambda}{a}$, band $\frac{\Delta y \approx \lambda L}{a}$, so $\frac{\Delta p_y \approx h}{a}$ and $\Delta y \cdot \Delta p_y \approx h$

standing-wave route

{chain} 1 $n\left(\frac{\lambda}{2}\right) = L \to \lambda_n = 2\frac{L}{n} \to p = \frac{h}{\lambda_n} \to E_n = \frac{p^2}{2}m = n^2\frac{h^2}{8mL^2}$

Week 5 lecture 1

the Schrödinger equation; operators and expectation values; the infinite square well

Schrödinger equation

A345 $i\hbar\left(\frac{\partial\Psi}{\partial t}\right) = -\left(\frac{\hbar^2}{2m}\right)\left(\frac{\partial^2\Psi}{\partial x^2}\right) + V(x)\Psi$ time-dependent Schrödinger equation *(Schrödinger equation)*

A346 $-\left(\frac{\hbar^2}{2m}\right)\left(\frac{d^2\psi}{dx^2}\right) + V(x)\psi = E\psi$ time-independent Schrödinger equation *(Schrödinger equation)*

A347 $f(t) = e^{-\frac{iEt}{\hbar}}$ separated time factor *(Schrödinger equation)*

A348 $\hat{p} = -\frac{i\hbar\partial}{\partial x}$ momentum operator *(Schrödinger equation)*

A349 $\hat{K} = -\left(\frac{\hbar^2}{2m}\right)\left(\frac{\partial^2}{\partial x^2}\right)$ kinetic energy operator *(Schrödinger equation)*

A350 $\hat{H} = -\left(\frac{\hbar^2}{2m}\right)\left(\frac{\partial^2}{\partial x^2}\right) + V(x)$ Hamiltonian in position representation *(Schrödinger equation)*

A351 $\frac{i\hbar\partial\Psi}{\partial t} = \hat{H}\Psi$ and $\hat{H}\psi = E\psi$ both equations as operator statements *(Schrödinger equation)*

A352 $\hat{E} = \frac{i\hbar\partial}{\partial t}$ energy operator *(Schrödinger equation)*

A353 $[\hat{x},\hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$ position-momentum commutator *(Schrödinger equation)*

A354 $\Psi = \sum_n c_n\psi_n e^{-\frac{iE_n t}{\hbar}}$ general solution as eigenstate superposition *(Schrödinger equation)*

A355 $\hat{p}e^{ikx} = \hbar k e^{ikx}$ plane waves as momentum eigenstates *(Schrödinger equation)*

A356 $\tilde{\psi}(k) = \left(\frac{1}{\sqrt{2\pi}}\right)\int\psi(x)e^{-ikx}\,\mathrm{d}x$ Fourier transform to momentum representation *(Schrödinger equation)*

A357 $\hbar\omega = \hbar^2\frac{k^2}{2}m$ free-plane-wave dispersion *(Schrödinger equation)*

A358 $E = \hbar\omega$ and $p = \hbar k$ matter-wave identifications *(Schrödinger equation)*

A359 $\psi_n = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)$, $E_n = n^2\frac{h^2}{8mL^2}$ infinite-well example quoted here *(Schrödinger equation)*

expectation value (quantum mechanics)

A360 $\langle A\rangle = \int\Psi^*\hat{A}\Psi\,\mathrm{d}x$ general expectation value definition *(expectation value (quantum mechanics))*

A361 $\langle x\rangle = \int x\,|\,\Psi(x,t)|^2\,\mathrm{d}x$ mean position *(expectation value (quantum mechanics))*

A362 $\langle p\rangle = \int\Psi^*\left(-\frac{i\hbar\partial}{\partial x}\right)\Psi\,\mathrm{d}x$ mean momentum *(expectation value (quantum mechanics))*

A363 $\langle p^2\rangle = \int\Psi^*\left(-\frac{\hbar^2\partial^2}{\partial x^2}\right)\Psi\,\mathrm{d}x$ mean-square momentum *(expectation value (quantum mechanics))*

A364 $\langle E\rangle = \int\Psi^*\left(\frac{i\hbar\partial}{\partial t}\right)\Psi\,\mathrm{d}x$ mean energy *(expectation value (quantum mechanics))*

A365 $\sigma_x^2 = \langle x^2\rangle - \langle x\rangle^2$, $\Delta x = \sigma_x$ variance and spread from moments *(expectation value (quantum mechanics))*

A366 $\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$ uncertainty bound on the spreads *(expectation value (quantum mechanics))*

A367 $\psi_1 = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)$, $\langle x\rangle = \frac{L}{2}$, $\langle p\rangle = 0$ box ground-state averages *(expectation value (quantum mechanics))*

A368 $\langle p^2\rangle = 2mE_1 = \frac{h^2}{4L^2}$, $\Delta p = \frac{h}{2L}$ ground-state momentum spread *(expectation value (quantum mechanics))*

A369 $\langle x^2\rangle = L^2\left(\left(\frac{1}{3}\right) - \left(\frac{1}{2\pi^2}\right)\right)$ ground-state mean-square position *(expectation value (quantum mechanics))*

A370 $\Delta x = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}} \approx 0.18L$ ground-state position spread *(expectation value (quantum mechanics))*

A371 $\Delta x \cdot \Delta p \approx 0.57\hbar$ box ground state above the floor *(expectation value (quantum mechanics))*

Derivation chains

ground-state ⟨x²⟩ double integration by parts

{chain} 1 $\sin^2\left(\frac{\pi x}{L}\right) = \left(\frac{1}{2}\right)\left(1 - \cos\left(\frac{2\pi x}{L}\right)\right)$ splits $\langle x^2\rangle = \left(\frac{1}{L}\right)\int_0^L x^2\,\mathrm{d}x - \left(\frac{1}{L}\right)I$ with $I = \int_0^L x^2\cos\left(\frac{2\pi x}{L}\right)\mathrm{d}x$; first integral is $\frac{L^2}{3}$

{chain} 2 $I = \left[x^2\cdot\left(\frac{L}{2\pi}\right)\sin\left(\frac{2\pi x}{L}\right)\right]_0^L - \left(\frac{L}{\pi}\right)\int_0^L x\sin\left(\frac{2\pi x}{L}\right)\mathrm{d}x$; boundary term vanishes since $\sin(0) = \sin(2\pi) = 0$

{chain} 3 $\int_0^L x\sin\left(\frac{2\pi x}{L}\right)\mathrm{d}x = \left[-\left(\frac{xL}{2\pi}\right)\cos\left(\frac{2\pi x}{L}\right)\right]_0^L + \left(\frac{L}{2\pi}\right)\int_0^L\cos\left(\frac{2\pi x}{L}\right)\mathrm{d}x$; cosine integral covers a full period and vanishes; boundary term leaves $-\frac{L^2}{2\pi}$

{chain} 4 $I = \frac{L^3}{2\pi^2}$, so $\langle x^2\rangle = L^2\left(\left(\frac{1}{3}\right) - \left(\frac{1}{2\pi^2}\right)\right)$; with $\langle x\rangle = \frac{L}{2}$ gives $\Delta x = L\sqrt{\frac{1}{12} - \frac{1}{2\pi^2}} \approx 0.18L$

{chain} 5 $\Delta x \cdot \Delta p \approx 0.57\hbar$, above the $\frac{\hbar}{2}$ floor since only a Gaussian saturates

⟨p⟩ = 0 boundary-term argument

{chain} 1 real Ψ means $\Psi^* = \Psi$ and $\frac{\Psi\partial\Psi}{\partial x} = \left(\frac{1}{2}\right)\partial\frac{\Psi^2}{\partial}x$, a perfect derivative

{chain} 2 $\langle p\rangle = -i\hbar\int\Psi\left(\frac{\partial\Psi}{\partial x}\right)\mathrm{d}x = -\left(\frac{i\hbar}{2}\right)\left[\Psi^2\right]_{-\infty}^{+\infty} = 0$ since Ψ falls to zero at both ends

{chain} 3 for complex Ψ, $\Psi^*\frac{\partial\Psi}{\partial x}$ is not a total derivative, so $\langle p\rangle$ can be nonzero

free-particle plane waves

{chain} 1 $\psi = e^{i(kx-\omega t)}$ into $V = 0$ equation gives $i\hbar(-i\omega)\psi = -\left(\frac{\hbar^2}{2m}\right)(ik)^2\psi$

{chain} 2 reduces to $\hbar\omega = \hbar^2\frac{k^2}{2}m$; with $E = \hbar\omega$, $p = \hbar k$ recovers $E = \frac{p^2}{2}m$, $\lambda = \frac{\hbar}{p}$

separation of variables

{chain} 1 try $\Psi(x,t) = \psi(x)f(t)$ and substitute into the time-dependent equation

{chain} 2 divide by ψf: $i\hbar\left(\frac{1}{f}\right)\left(\frac{\mathrm{d}f}{\mathrm{d}t}\right) = E$ and $-\left(\frac{\hbar^2}{2m}\right)\left(\frac{1}{\psi}\right)\left(\frac{d^2\psi}{dx^2}\right) + V = E$, each equal to constant E

{chain} 3 $i\hbar\left(\frac{1}{f}\right)\left(\frac{\mathrm{d}f}{\mathrm{d}t}\right) = E$ integrates to $f(t) = e^{-\frac{iEt}{\hbar}}$

{chain} 4 space part is the eigenvalue problem $-\left(\frac{\hbar^2}{2m}\right)\psi'' + V\psi = E\psi$ with quantised E

Week 5 lecture 2

the finite square well and penetration depth; three-dimensional wells; the harmonic oscillator; tunnelling and the STM

finite potential well

A372 $\alpha = \frac{\sqrt{2m(V_0-E)}}{\hbar}$ decay constant outside finite well *(finite potential well)*

A373 $\psi_{\mathrm{I}} = Ae^{\alpha x}$, $\psi_{\mathrm{III}} = Be^{-\alpha x}$ decaying wall solutions *(finite potential well)*

A374 $\psi_{\mathrm{II}} = Ce^{ikx} + De^{-ikx}$, $k = \frac{\sqrt{2mE}}{\hbar}$ oscillating interior solution *(finite potential well)*

A375 $\left(\frac{d^2\psi}{dx^2}\right) = \left(\frac{2m(V_0-E)}{\hbar^2}\right)\psi$ in walls rearranged wall-region equation *(finite potential well)*

A376 $\delta = \frac{1}{\alpha} = \frac{\hbar}{\sqrt{2m(V_0-E)}}$ penetration depth of amplitude *(finite potential well)*

A377 $|\psi|^2 \sim e^{-2\frac{x}{\delta}}$, density e-folding length $\frac{\delta}{2}$ probability falls twice as fast *(finite potential well)*

quantum harmonic oscillator

A378 $E_n = \left(n + \left(\frac{1}{2}\right)\right)\hbar\omega$, $\omega = \sqrt{\frac{k}{m}}$ oscillator level ladder *(quantum harmonic oscillator)*

A379 $E_0 = \left(\frac{1}{2}\right)\hbar\omega$ zero-point energy *(quantum harmonic oscillator)*

A380 $V(x) \approx V_0 + \left(\frac{1}{2}\right)k(x-x_0)^2$, $k = V_2$ harmonic approximation at minimum *(quantum harmonic oscillator)*

A381 $\left(\frac{d^2\psi}{dx^2}\right) + \left(\left(\frac{2mE}{\hbar^2}\right) - \left(\frac{m^2\omega^2}{\hbar^2}\right)x^2\right)\psi = 0$ Schrödinger equation for parabolic V *(quantum harmonic oscillator)*

A382 $\xi = \sqrt{\frac{m\omega}{\hbar}}x$, $\epsilon = 2\frac{E}{\hbar\omega}$, $\psi'' + (\epsilon - \xi^2)\psi = 0$ rescaled oscillator equation *(quantum harmonic oscillator)*

A383 $\psi = H(\xi)e^{-\frac{\xi^2}{2}}$ Gaussian factored form *(quantum harmonic oscillator)*

A384 $\psi' = (H' - \xi H)e^{-\frac{\xi^2}{2}}$, $\psi'' = (H'' - 2\xi H' + (\xi^2 - 1)H)e^{-\frac{\xi^2}{2}}$ factored derivatives *(quantum harmonic oscillator)*

A385 $H'' - 2\xi H' + (\epsilon - 1)H = 0$ Hermite equation *(quantum harmonic oscillator)*

A386 $h_{j+2} = (2j + 1 - \epsilon)\frac{h_j}{(j+1)(j+2)}$ Hermite-series recurrence *(quantum harmonic oscillator)*

A387 $\epsilon = 2n + 1$ from $h_{n+2} = 0$ with $h_n \neq 0$ series termination condition *(quantum harmonic oscillator)*

A388 $\psi_{n(x)} = H_{n(x)}e^{-\frac{m\omega x^2}{2\hbar}}$ Hermite times Gaussian eigenfunctions *(quantum harmonic oscillator)*

A389 $E(\Delta x) = \frac{\hbar^2}{8m(\Delta x)^2} + \left(\frac{1}{2}\right)m\omega^2(\Delta x)^2$ uncertainty energy bound *(quantum harmonic oscillator)*

A390 $\Delta x = \sqrt{\frac{\hbar}{2m\omega}}$ optimal localisation width *(quantum harmonic oscillator)*

A391 $p_{\mathrm{cl}}(x) = \frac{1}{\pi\sqrt{A^2 - x^2}}$ classical oscillator probability density *(quantum harmonic oscillator)*

A392 large-n envelope $\frac{\propto 1}{\sqrt{x_T^2 - x^2}}$ quantum density classical limit *(quantum harmonic oscillator)*

quantum tunnelling

A393 $\kappa = \frac{\sqrt{2m(V_0-E)}}{\hbar}$ decay constant inside barrier *(quantum tunnelling)*

A394 $\psi_{\mathrm{II}} = Ce^{\kappa x} + De^{-\kappa x}$ barrier-interior solution *(quantum tunnelling)*

A395 $k = \frac{\sqrt{2mE}}{\hbar}$ wave number in free regions *(quantum tunnelling)*

A396 $\psi_{\mathrm{I}} = e^{ikx} + re^{-ikx}$ region I incident plus reflected *(quantum tunnelling)*

A397 $\psi_{\mathrm{III}} = te^{ikx}$ region III transmitted only *(quantum tunnelling)*

A398 $1 + r = A + B$ ψ match at x = 0 *(quantum tunnelling)*

A399 $ik(1 - r) = \kappa(-A + B)$ ψ' match at x = 0 *(quantum tunnelling)*

A400 $Ae^{-\kappa L} + Be^{\kappa L} = te^{ikL}$ ψ match at x = L *(quantum tunnelling)*

A401 $\kappa\left(-Ae^{-\kappa L} + Be^{\kappa L}\right) = ikte^{ikL}$ ψ' match at x = L *(quantum tunnelling)*

A402 $t = \left(\frac{4k\kappa e^{-ikL}}{(k+i\kappa)^2 e^{-\kappa L} - (k-i\kappa)^2 e^{\kappa L}}\right)$ transmitted amplitude *(quantum tunnelling)*

A403 $T = |t|^2$ transmission probability definition *(quantum tunnelling)*

A404 $T = \left[1 + \left(\frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0-E)}\right)\right]^{-1}$ exact rectangular-barrier transmission *(quantum tunnelling)*

A405 $T \approx 16\left(\frac{E}{V_0}\right)\left(1 - \left(\frac{E}{V_0}\right)\right)e^{-2\kappa L}$ thick-barrier approximation *(quantum tunnelling)*

A406 $V_0 - E = mg(H - h) \approx 10^4\,\mathrm{J}$ (lion) macroscopic barrier excess energy *(quantum tunnelling)*

A407 $\kappa \approx 3 \times 10^{37}\,\mathrm{m}^{-1}$ (lion) macroscopic decay constant *(quantum tunnelling)*

A408 $\kappa L \approx 1.5 \times 10^{37}$, T underflows to zero (lion) lion tunnelling suppression *(quantum tunnelling)*

A409 $V_0 - E = 4\ \mathrm{eV} \approx 6.4 \times 10^{-19}\mathrm{J}$ (electron) electron barrier excess energy *(quantum tunnelling)*

A410 $\kappa \approx 1.1 \times 10^{10}\,\mathrm{m}^{-1}$ (electron) electron decay constant *(quantum tunnelling)*

A411 $\kappa L \approx 11$, $e^{-2\kappa L} \approx 10^{-10}$, prefactor ≈ 3, $T \sim 10^{-9}$ electron tunnelling estimate *(quantum tunnelling)*

scanning tunnelling microscope (quantum tunnelling): hover sharp tip within d < 1 nm of conducting surface, apply bias V
Measure: tunnelling current I. Determines: surface topography via I ∝ V e^{-2κd}.

Derivation chains

QHO series to termination E_n = (n+½)ħω

{chain} 1 $V = \left(\frac{1}{2}\right)kx^2$, $\omega = \sqrt{\frac{k}{m}}$ into TISE gives $\left(\frac{d^2\psi}{dx^2}\right) + \left(\left(\frac{2mE}{\hbar^2}\right) - \left(\frac{m^2\omega^2}{\hbar^2}\right)x^2\right)\psi = 0$

{chain} 2 $\xi = \sqrt{\frac{m\omega}{\hbar}}x$, $\epsilon = 2\frac{E}{\hbar\omega}$ turn it into $\psi'' + (\epsilon - \xi^2)\psi = 0$

{chain} 3 normalisability demands $\psi = H(\xi)e^{-\frac{\xi^2}{2}}$; substituting its derivatives cancels the ξ^2 terms and leaves $H'' - 2\xi H' + (\epsilon - 1)H = 0$

{chain} 4 $H = \sum_j h_j\xi^j$ in the Hermite equation gives $h_{j+2} = (2j + 1 - \epsilon)\frac{h_j}{(j+1)(j+2)}$

{chain} 5 $h_{n+2} = 0$ with $h_n \neq 0$ forces $\epsilon = 2n + 1$, hence $E_n = \left(n + \left(\frac{1}{2}\right)\right)\hbar\omega$

{chain} 6 any other ϵ leaves $\frac{h_{j+2}}{h_j} \to 1$, so $H \sim e^{+\xi^2}$ and $\psi \sim e^{+\frac{\xi^2}{2}}$, not normalisable

zero-point minimisation

{chain} 1 $\langle p\rangle = 0$, $\langle p^2\rangle = (\Delta p)^2$, $\langle V\rangle = \left(\frac{1}{2}\right)m\omega^2(\Delta x)^2$ give $E \geq \frac{\hbar^2}{8m(\Delta x)^2} + \left(\frac{1}{2}\right)m\omega^2(\Delta x)^2$

{chain} 2 $\frac{\mathrm{d}E}{\mathrm{d}\Delta x} = -\frac{\hbar^2}{4m(\Delta x)^3} + m\omega^2\Delta x = 0$ gives $\Delta x^4 = \frac{\hbar^2}{4m^2\omega^2}$, so $\Delta x = \sqrt{\frac{\hbar}{2m\omega}}$

{chain} 3 at $\Delta x^2 = \frac{\hbar}{2m\omega}$ the confinement and potential terms each equal $\frac{\hbar\omega}{4}$, so the minimum is $E = \frac{\hbar\omega}{2}$

{chain} 4 the true ground state saturates the uncertainty bound, so the estimate is exact for it

tunnelling T matching chain

{chain} 1 region I $\psi = e^{ikx} + re^{-ikx}$; region II $\psi = Ae^{-\kappa x} + Be^{\kappa x}$; region III $\psi = te^{ikx}$, nothing incident from the right

{chain} 2 ψ and ψ' continuous at $x = 0$ and $x = L$ give $1 + r = A + B$; $ik(1 - r) = \kappa(-A + B)$; $Ae^{-\kappa L} + Be^{\kappa L} = te^{ikL}$; $\kappa\left(-Ae^{-\kappa L} + Be^{\kappa L}\right) = ikte^{ikL}$

{chain} 3 eliminating r, A, B leaves $t = \left(\frac{4k\kappa e^{-ikL}}{(k+i\kappa)^2 e^{-\kappa L} - (k-i\kappa)^2 e^{\kappa L}}\right)$

{chain} 4 $T = |t|^2$ with $e^{\kappa L} - e^{-\kappa L} = 2\sinh(\kappa L)$ reduces the denominator to $T = \left[1 + \left(\frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0-E)}\right)\right]^{-1}$

{chain} 5 $\kappa L \gg 1$ makes sinh exponential, giving $T \approx 16\left(\frac{E}{V_0}\right)\left(1 - \left(\frac{E}{V_0}\right)\right)e^{-2\kappa L}$; amplitude $e^{-\kappa L}$ squared carries $e^{-2\kappa L}$

{chain} 6 $V_0 - E = mg(H - h) \approx 10^4$ J $\to \kappa \approx 3 \times 10^{37}\,\mathrm{m}^{-1} \to \kappa L \approx 1.5 \times 10^{37} \to e^{-2\kappa L}$ underflows

{chain} 7 $V_0 - E \approx 6.4 \times 10^{-19}$ J $\to \kappa \approx 1.1 \times 10^{10}\,\mathrm{m}^{-1} \to \kappa L \approx 11 \to$ prefactor $\approx 3 \to T \sim 10^{-9}$

STM sensitivity chain

{chain} 1 wide-gap barrier of height φ gives $T_{\mathrm{gap}} \approx e^{-2\kappa d}$ with $\kappa = \frac{\sqrt{2m\varphi}}{\hbar}$

{chain} 2 incident flux (carrying bias V) times transmission gives $I \propto Ve^{-2\kappa d}$

{chain} 3 $\frac{\mathrm{d}I}{I} = -2\kappa dd$; with $\varphi \approx 5$ eV, $\kappa \approx 1.15\text{Å}^{-1}$, $2\kappa \approx 2.3\text{Å}^{-1}$

{chain} 4 each extra ångström multiplies I by $e^{-2.3} \approx 0.1$, near an order of magnitude, giving sub-ångström distance tracking